



Coupole

Reference manual

Version 0.2.0 — 9 October 2026

Free toolbox for students of the « Explorer et Comprendre l'Univers » diploma
(Observatoire de Paris).

OHP image bank · image quality · spectra and series · sites and times ·
cosmology.

Contents

1	Overview	2
2	Installation	3
2.1	All-inclusive packages	3
2.2	With the computer's Python	3
2.3	pip and pipx	4
2.4	Updates, uninstallation, data	4
3	Getting started	5
3.1	Choosing a file or a folder, folder on a NAS	5
3.2	Appearance	6
3.3	Saved settings	6
4	OHP image bank module	8
4.1	The bank	8
4.2	Catalogue	8
4.3	Processing	9
4.4	Download everything, log, reorganise	10
4.5	Output folder on a network share	10
4.6	New images in the bank	11
4.7	Astrometric solution	11
4.8	Fixed headers	12
4.9	Stacks	12
4.10	Opening an image	14
4.11	Anomalies	15
4.12	Sky map	16
4.13	Online record (optional)	16
5	Observatory archives	18
5.1	Searching	19
5.2	Downloading and preparing	19
5.3	Aligning and colouring	20
5.4	Command line	20
6	ASTAP (optional)	22
7	Output formats	24
8	Image quality (optional)	26
9	Spectra and series	28
10	Sites and times	29
11	Cosmology	30
12	Computer and performance	33
13	Command line	35
14	Evolution	43
15	Incident reports and privacy	44

16 Writing a module	45
17 Troubleshooting	46
Sources	47

CHAPTER 1

Overview

Coupole is a free application (GPL-3.0) written in Python 3.10+, running on Windows, macOS and Linux, with a graphical interface (PyQt6) and a complete command line. Every displayed text exists in French and English; the system language is detected and can be forced (*Language* menu, Preferences, or `-lang fr|en`).

Module	Purpose
OHP image bank	Inventory, catalogue, download, astrometric check, fixed headers, conversion, stackable sets, anomalies, sky map.
Observatory archives	Public images from Hubble, JWST, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno: search, volume, download, extraction, alignment, colour.
Image quality	Optional. FWHM, ellipticity, background, noise, gradient, saturation, trails, sampling.
Spectra and series	Reading and plotting 1D data (radio spectra, light curves, tables).
Cosmology	From redshift to distances, ages, volume, distance modulus, scale (Planck 2018 and others).
Sites and times	Site database, OpenStreetMap map, UTC and local time.
My computer	Hardware diagnosis; minimal example module.

Principles: all long work runs off the GUI thread (the interface stays responsive, with progress and cancellation); an interrupted run resumes where it stopped; nothing is deleted without a trace; no data is sent without consent; optional components (ASTAP, SEP, CuPy) may be missing.

CHAPTER 2

Installation

2.1 All-inclusive packages

The packages bundle Python and every library (PyQt6, numpy, astropy, zstandard, lz4, SEP): nothing else to install. They update themselves. ASTAP stays optional and is not included.

System	File	Installation
Windows 10/11 64-bit	Coupole-Setup-x.y.z.exe	Double-click. SmartScreen warning at first launch (unsigned installer): « More info », « Run anyway ». Installed in the user profile, no administrator rights.
macOS Apple Silicon	Coupole-x.y.z-macos-arm64.dmg	Drag Coupole into Applications. First launch: right click, « Open ».
macOS Intel	Coupole-x.y.z-macos-x86_64.dmg	Same.
Ubuntu / Debian x86_64, ARM64	coupole_x.y.z_amd64.deb	<pre>sudo apt install ./coupole_x.y.z_amd64.deb (_arm64.deb on ARM): apt installs the Qt system libraries; coupole command, menu entry. Ubuntu 22.04 or newer, Debian 12 or newer. Uninstall: sudo apt remove coupole.</pre>
Other Linux x86_64 / ARM64	Coupole-x.y.z-linux-*.tar.gz	<pre>Unpack, run installer.sh (copied into ~/.local/share/coupole, menu entry, coupole command through a link ~/.local/bin/coupole; if that folder is not in the PATH, the installer says so and gives the line to add). The Coupole.sh launcher follows symbolic links (links to links included).</pre>

2.2 With the computer's Python

The scripts in the source folder look for Python 3.10 or newer, create a virtual environment, install the dependencies and create launchers. One question only: Enter to confirm (-oui / -Oui to skip it).

>_ Command

```
sh install.sh # Linux, macOS
install.bat # Windows (double-click; runs install.ps1)
```

Without Python, or without the venv module (Debian: python3-venv package), the script says so and explains how to get it. Locations: Linux ~/.local/share/coupole/venv, launchers in ~/.local/bin; macOS ~/Library/Application Support/Coupole/venv and ~/Applications/Coupole.app; Windows %LOCALAPPDATA%\Programs\CoupolePython, Desktop and Start menu shortcuts.

2.3 pip and pipx

>_ Command

```
pipx install git+https://github.com/ARP273-ROSE/coupole
pip install . # from the sources; pip install ".[qualite]" adds SEP
python -m coupole # interface; coupole -help: command line
```

Extras: `qualite` (SEP), `validation` (lxml, xisf: XSD validation and second reader), `gpu` (CuPy), `test` (pytest).

2.4 Updates, uninstallation, data

At startup (can be disabled) the packages check the latest release on GitHub and offer to install it: only the code archive (`coupole-app-x.y.z.zip`, about 1 MB) is downloaded and unpacked by the application, the previous version being restored on failure. A pip/pipx installation shows the command to run. The `.deb` package does not update through the archive: its files (`/opt/coupole`) belong to `dpkg`; Coupole announces the new version, gives the address of the new `.deb` and the `apt` command to run (*Download the package* button). Settings, cache and reports live outside the installation folder and survive updates as well as removal:

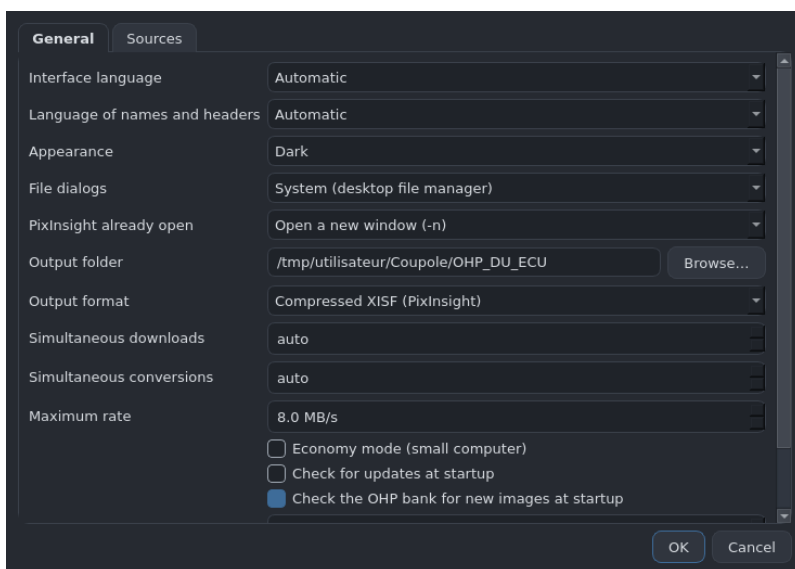
Windows	%LOCALAPPDATA%\Coupole
macOS	~/Library/Application Support/Coupole (cache: ~/Library/Caches/Coupole)
Linux	~/.config/coupole (cache: ~/.cache/coupole)
All	COUPOLE_HOME variable: replaces both (portable installations, tests)

CHAPTER 3

Getting started

The window has a module bar on the left (Ctrl+1 to Ctrl+9) and the current module on the right. Every interactive element has a tooltip. *Help* menu: help on the screen shown (F1), this manual (Shift+F1), keyboard shortcuts, report a problem, incident reports, check for updates, About (version, licence, detected configuration).

F1	help on the screen
Shift+F1	PDF manual
Ctrl+1 ... Ctrl+9	module 1 to 9
Ctrl+,	Preferences
Ctrl+Shift+A	ASTAP assistant
Ctrl+Shift+D	dark / light theme
Ctrl+R	OHP bank: refresh the inventory
Ctrl+Q	quit



Preferences (Ctrl+,): interface language; language of folder names, object names and FITS comments; appearance (section 3.2); file dialogs (section 3.1); PixInsight already open (section 4.10); output folder; format; parallelism; maximum rate; economy mode; update check; check of the bank for new images at startup and its maximum frequency (section 4.6). The *Sources* tab lists the service addresses (chapter 14). An unreadable settings file is set aside (suffix `.corrompu-...`) and the defaults take over: the application always starts.

3.1 Choosing a file or a folder, folder on a NAS

The *Browse* buttons open the system file manager, with its places, bookmarks and network shares: Explorer on Windows, Finder (NSOpenPanel) on macOS, and on Linux the desktop's own through the **XDG portal** (`xdg-desktop-portal`): Dolphin on KDE Plasma, Files (Nautilus) on GNOME. The standalone package bundles its own Qt library, which cannot load the desktop plugin installed on the system: up to 0.1.7 it fell back to Qt's generic dialog, in English and without the system places. At startup, unless the user already set `QT_QPA_PLATFORMTHEME`, Coupole checks (waiting at most one second) whether the portal is installed or running with a backend able to choose files, and uses it; otherwise, on a GTK desktop (GNOME, Cinnamon, Xfce, MATE...), the GTK dialog; otherwise Qt's dialog, **in the interface language**, its sidebar completed with the home folders, `/media`, `/mnt`, `/run/media/$USER`, the shares opened in the file manager (`/run/user/<uid>/gvfs/...`) and the cifs, nfs and sshfs mounts found in `/proc/mounts`.

Preferences > *File dialogs: System* (default) or *Qt*, the latter only if the system dialog does not open or misbehaves (broken portal). On Linux, switching back to *System* takes effect at next start. All texts provided by Qt (standard buttons, context menus of text fields, fallback dialog) follow the interface language and change with it.

Output folder on a NAS.

- *Linux*: first open the share in Dolphin or Files (`smb://server/share`); it shows up under `/run/user/<uid>/gvfs/smb-share:server=server,share=share` (GNOME; KDE with the `kio-fuse` package, without which an unmounted `smb://` share cannot be picked by an application), or mount the share with `cifs` or `nfs` (`/mnt/partage`, advised for large volumes; default options, no need for `nobrl`: section 4.5);
- *macOS*: Finder > Go > Connect to Server (Cmd+K), `smb://server/share`, then pick the volume under `/Volumes`;
- *Windows*: network drive (Z:) or UNC path `\\server\share`, which can also be typed into the folder field.

These paths are accepted everywhere: state database read read-only (SQLite URI of a UNC path written with an empty authority), free space, write test, Windows long paths (`\\?\UNC\...`), wrapped display; a share is recognised as such (UNC, network drive, `cifs`, `nfs`, `sshfs`, `gvfs`, `kio-fuse`) and the Quality module limits its parallel reads there.

3.2 Appearance

Coupole has its own theme, independent of the computer's (a system in dark mode no longer mixes its colours with the application's: Fusion style, complete palette, fixed font size, WCAG contrast ≥ 4.5 for text, fields, buttons, tooltips and selection). The **dark** theme is the default; the light theme remains available. It is changed through the *View > Appearance* menu (checkable *Light / Dark* entries, Ctrl+Shift+D to toggle) or in the Preferences: the same setting, applied at once and saved; both stay in sync. A user who had chosen the light theme keeps it: the saved setting takes precedence over the default. The screenshots of this manual are taken in the default theme.

3.3 Saved settings

Each time it starts, Coupole shows the screen as it was left. Kept:

- **window**: size, position, screen, maximised state; size of every dialog (Preferences and its tab, ASTAP assistant, About, report, help, object correction, new site);
- **layout**: module shown, tab of the OHP bank, splitter positions, width and order of the columns, sort column and direction of every table;
- **OHP bank**: search, type, telescope, *New*, *To check*, *Missing only* and *Without doubtful dates* boxes, night and filter, chosen objects (restored as soon as the inventory is loaded), anomaly and sky-map filters; Processing tab: output folder *as soon as it is changed*, format, language of names, *Keep duplicates*, *Keep FITS*, ASTAP mode, *Check quality*;
- **Image quality**: analysed folder, sample mode and number of images per batch;
- **Spectra and series**: folder of the open dialog and *Recent* menu (last 10 files; a file that has disappeared is greyed out); chosen axis;
- **Cosmology**: parameter set, custom parameters (H_0 , Ω_m , kept even when going back to Planck), Ω_k , current z , SH0ES option, scale of the curves (logarithmic or linear);
- **Sites and times**: chosen site, zoom and centre of the world map, *Online map*;
- **file dialogs**: each one opens again in the last folder used (CSV export, reorganisation, ASTAP, report...).

The layout lives in `interface.json`, next to `reglages.json` (section 2.4); processing choices stay in `reglages.json`. Writing is *deferred and grouped*: at most once every 2 s and once on closing, atomic (temporary file then replacement); typing in a field never writes to disk. Tolerant reading: a missing value, a wrong type or an out-of-range value leaves the default; an unreadable file is set aside (`.corrompu-...`). Safeguards: an unplugged screen or a position outside the screens centres the window again, a size larger than the screen is brought back onto it (as at 150 or 200 %). A folder kept on a network share that is absent never blocks start-up: its existence is checked in the background, and the dialog opens in the default folder until it is confirmed. Nothing that would be dangerous to replay is kept: an interrupted processing never restarts on its own (it resumes when you run it again).

To go back to the original: Preferences > *Reset the layout* (immediate; language, theme, output folder and the other settings do not change), or `coupole -reinitialiser-interface (-reset-interface)`.

CHAPTER 4

OHP image bank module

4.1 The bank

« OHP student observations » gathers the images taken by the diploma's students at the Observatoire de Haute-Provence: 7,989 FITS images on 8 October 2026, 166 objects under 194 names, 60 nights from 2015 to 2026, 78.0 GB; 1.20 m telescope (T120, 7,419 images, 1024 pixels of 0.77", 13.1' field) and the 50 cm IRIS robotic telescope (570 images, 2048 pixels of 0.68" or 4096 pixels of 0.46", 23' and 31' fields). The images are already calibrated (dark, flat) and astrometrically solved: **they must not be recalibrated**.

4.2 Catalogue

The screenshot shows the 'Catalogue' tab of the OHP image bank software. The interface includes a sidebar with navigation options: 'OHP image bank', 'Observatory archives', 'Image quality', 'Spectra and series', 'Cosmology', 'Sites and times', and 'My computer'. The main window displays a table of objects with columns for type, object name, and owned status. A summary bar indicates 7,989 images (144 duplicates), 166 objects, 60 nights, and 78.03 GB. A table on the right shows a list of objects with their UTC date, site time, night, and telescope. The bottom status bar shows 'Ready' and a progress bar for downloading images.

type	object	owned
Planets	Saturn	28 / 56
Planets	Uranus	17 / 34
Trans-Neptunian objec...	(134340) Pluto	0 / 280
Trans-Neptunian objec...	(136108) Haumea	23 / 23
Trans-Neptunian objec...	(136199) Eris	10 / 21
Trans-Neptunian objec...	(145452) Ritona	26 / 52
Trans-Neptunian objec...	(50000) Quaoar	0 / 32
Trans-Neptunian objec...	(55636) 2002 TX300	0 / 23
Planetary nebulae	M27	136 / 273
Planetary nebulae	M57	126 / 252
Planetary nebulae	M76	0 / 14
Planetary nebulae	NGC 6210	5 / 5
Planetary nebulae	NGC 6543	88 / 88
Planetary nebulae	NGC 6826	14 / 14
Planetary nebulae	NGC 7026	152 / 153
Planetary nebulae	NGC 7662	30 / 30
Nebulae and remnants	Barnard 68	64 / 129
Nebulae and remnants	IC 1396	31 / 63
Nebulae and remnants	IC 434	10 / 10
Nebulae and remnants	IC 5070	24 / 48
Nebulae and remnants	IC 5146	0 / 93
Nebulae and remnants	M16	117 / 235
Nebulae and remnants	M17	0 / 143
Nebulae and remnants	M20	123 / 123
Nebulae and remnants	M42	13 / 14
Nebulae and remnants	NGC 2237	3 / 6
Nebulae and remnants	NGC 6888	430 / 840

owned	UTC date	site time	night	tel.	
owned	2025-07-16 21:25:54	23:25:54 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:25:58	23:25:58 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:02	23:26:02 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:05	23:26:05 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:26:09	23:26:09 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:12	23:26:12 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:16	23:26:16 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:26:20	23:26:20 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:23	23:26:23 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:27	23:26:27 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:30	23:26:30 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:34	23:26:34 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:38	23:26:38 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:41	23:26:41 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:45	23:26:45 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:26:48	23:26:48 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:52	23:26:52 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:55	23:26:55 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:26:59	23:26:59 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:27:02	23:27:02 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:27:06	23:27:06 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:27:10	23:27:10 (UTC+0200)	2025-07-16	T120	lol
owned	2025-07-16 21:27:13	23:27:13 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:27:17	23:27:17 (UTC+0200)	2025-07-16	T120	lol
to download	2025-07-16 21:27:20	23:27:20 (UTC+0200)	2025-07-16	T120	lol

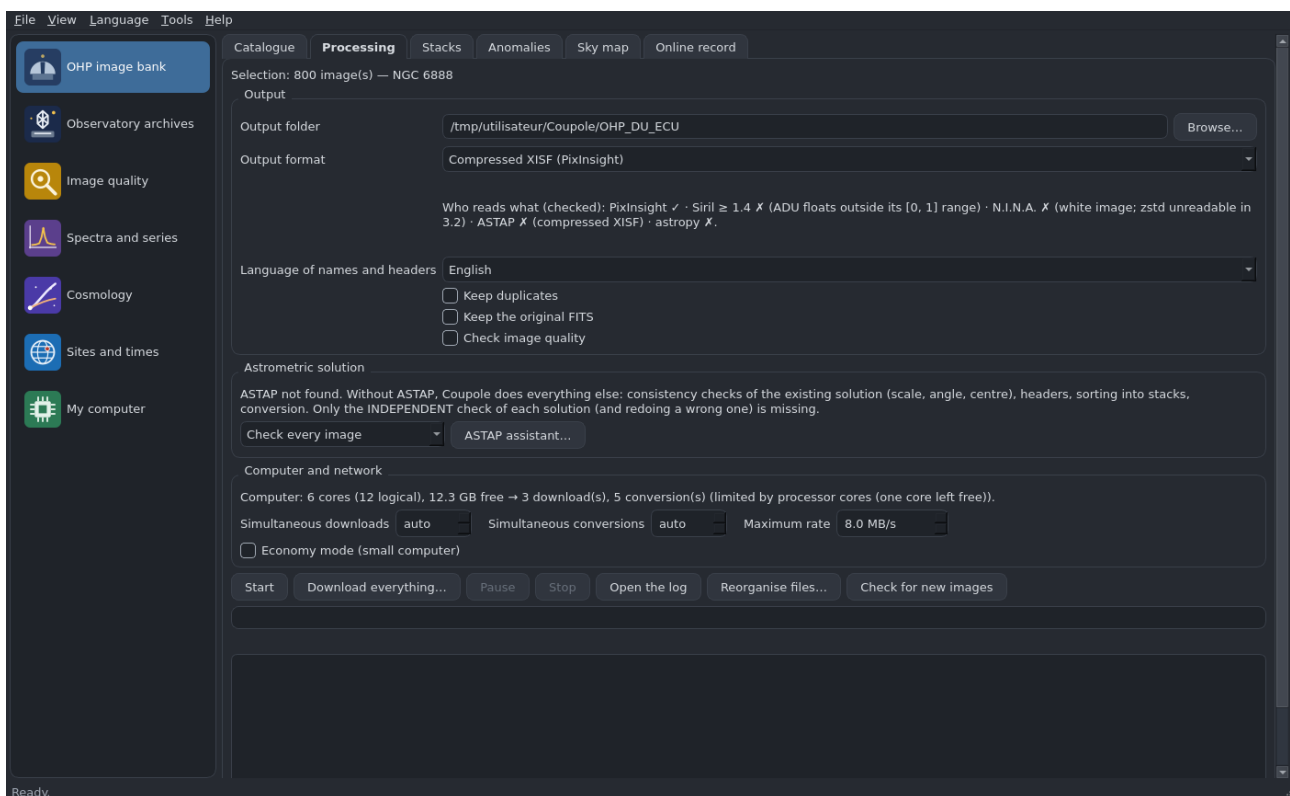
Ready. 300 missing image(s) out of 840 (430 already owned): 310 GB to download → about 1.16 GB of VHS. Exp. at destination: 735.98 GB

The inventory comes from the Observatory's public TAP service (<http://tap-ufe.obspm.fr/tap>, table `ivoa.obscore`, one synchronous ADQL query), kept in a cache; a snapshot is shipped for offline work. *Refresh the inventory* (Ctrl+R) queries the service again and flags new images and names (« new since ... »). Database names are grouped into canonical objects; selecting one or several objects lists their images on the right, filtered by night, filter and doubtful dates; this list is the selection to process. The estimate gives the volume to download, the expected output volume and the free space. Times are shown in UTC and in site time (hover: also the user's time).

What you already own. According to the output folder (`_traitement/etat.sqlite`, read again in the background when opening, after each run and whenever the folder changes, never blocking), the image list tells apart with a marker and a soft text colour (readable in both themes, contrast ≥ 4.5): **owned** (converted and sorted, green check), **duplicate left out** (same pixels as an image already converted, grey dot), **failed** (to retry, orange triangle) and **to download** (normal text, arrow); the marker's tooltip gives the status, the date and the local path of the file. The **object list** shows the same state, in the same colours: each row starts with a marker (green check: everything owned, left-out duplicates included; green half disc: partly; orange triangle: at least one failure; arrow: nothing downloaded), the object name takes the same colour, and the tooltip details it (« 12 owned / 12 · 0 to

download · 0 failed · 3 duplicate(s) left out »). The *owned* column (« 120 / 300 », mini bar) comes right after the name (a column order chosen by hand is kept); sorting it orders objects by state (failed, nothing, partly, complete). The legend, below both lists, applies to both. The inventory summary line adds the overall ownership (« 7,989 images (144 duplicates), 166 objects, 60 nights, 78.03 GB — owned: 7,625 (+ 220 duplicates left out), failed: 0, to download: 0 »). The *To download only* box hides fully owned objects and, in the list, the images already processed; an emptied list explains itself in its centre (« Everything is already downloaded in ... »), as does the image list while no object is selected (« Pick one or more objects... »; at first start, the first object is selected). The estimate counts only what is missing (« 180 missing images out of 300: 1.5 GB »). Changing the output folder (Preferences, field or *Browse* in the Processing tab) re-reads its ownership in the background at once: markers, column and summary follow, and the status bar says so (« Ownership recomputed: 7,625 images found in ... »). Command line: `coupole ohp inventory -missing [-dest FOLDER] [-json]`, and `possedee`, `statut_local`, `fichier_local` columns in the CSV of `coupole ohp list`.

4.3 Processing



For each image, duplicates excluded:

1. download: at most 4 at once (3 by default), total rate capped (8 MB/s by default), User-Agent naming Coupole, resume by HTTP Range requests, checks of size (server and inventory), multiple of 2,880 bytes and SIMPLE card;
2. reading of the FITS cards as written and of the pixels; SHA-1 fingerprint of the pixels (exact duplicates);
3. astrometric solution check (section 4.7);
4. fixed header (section 4.8);
5. writing in the chosen format, reading back and pixel-by-pixel comparison;
6. at the end: sorting into stacks, `LOT.txt`, `INDEX_LOTS.csv`, `_traitement/journal.csv`.

The state is kept in `_traitement/etat.sqlite`: *Stop* then *Start* (or running the same command again) resumes without redoing anything. A folder keeps the name language and format of its first

run. *Keep duplicates* also processes what would be left out; *Keep the original FITS* stores them in `_traitement/fits_origine`.

The chain is a *pipeline*: downloads (threads) and conversions (separate processes, one per available core) overlap, the queue of waiting FITS is bounded ($2 \times$ conversions + downloads; when free space at the destination is tight it is reduced automatically — never below conversions + 1 — and the log says so, rather than refusing the job), and the overall rate is that of the network or of the chosen cap (measured: 24 images of 8 MB at 8 MB/s in 24.5 s, i.e. the duration of the download alone; without a cap, 86 MB/s of FITS converted on 6 cores). *Pause* suspends feeding the pipeline (images in progress finish) and *Resume* restores it; *Stop* interrupts downloads and conversions without waiting (processes are terminated, ASTAP included: by signal on Linux and macOS, by a *job object* on Windows). At the end, a report sums up images, duration, converted, duplicates, failures (with their reason), volume written and stacks.

On a network share (NAS over SMB), where every access is a round trip, the state is committed at most once per second and `JOURNAL.txt` is written in batches (a power cut only redoes the last second); the final sorting rewrites only the `LOT.txt` files that changed and no longer walks the whole tree: running a processing again for one image no longer costs a minute of sorting. The state database, however, is never written directly on a share: section 4.5.

4.4 Download everything, log, reorganise

Download everything... takes the whole bank (about 8,000 images, 78 GB of FITS, 30 GB as XISF). On first use (or when no output folder is set), Coupole asks for the destination folder through a native dialog and proposes `Documents/Coupole/OHP_DU_ECU`; then it shows, **before any confirmation**, the volume to download, the expected output volume, the estimated duration at the rate cap (78 GB at 8 MB/s: 2 h 42) and the free space; it clearly refuses when space is short. Processing is the same as for any selection: resume after a cut or a close (per-image state in `_traitement/etat.sqlite`, already downloaded FITS found again), pause, end report. Command line: `coupole ohp all -yes`.

Open the log shows `_traitement/JOURNAL.txt`: one timestamped (UTC), bilingual line per event — session start and end (resume, pause), each image downloaded, converted or left out with its reason, each error, each name conflict; rotation at 5 MB (`JOURNAL-1.txt`, `JOURNAL-2.txt`). The tabular details stay in `journal.csv`.

Reorganise files... sorts into the stack tree images already converted by Coupole that live elsewhere or in an older layout: each file is recognised by its header (source address `OHP:Source:URL`, or HISTORY card « converted from ... »), attached to the inventory and **moved** (never copied); a name already taken by a foreign file gets a suffix and a log line; nothing is ever overwritten. Headers are read by several processes, progress is shown (« 1,200 / 7,625 headers read ») and *Stop* interrupts the reading (whatever is already recognised is sorted). Command line: `coupole ohp reorganise FOLDER -dest DEST`.

4.5 Output folder on a network share

Why. The state database (`_traitement/etat.sqlite`) is an SQLite database, which locks its file on every write. On Linux, an SMB share mounted by `cifs` with default options passes these locks on to the server as SMB byte-range locks, and writing fails: « database is locked » after one minute, 0-byte database (checked on a real Samba server; true of every version up to 0.1.9: downloading new images, *Download everything* or reorganising into a NAS folder failed). Read-only access and file copies do work.

What Coupole does (0.1.10). When the output folder is recognised as a share (`cifs`, `smb3`, `nfs`, `sshfs`, `gvfs`, `kio-fuse` on Linux; `smbfs`, `afpfs`, `nfs` on macOS; UNC path or network drive on Windows) or when SQLite cannot write there within 3 s (unrecognised share, another program), it works on a **local working database** in the user's cache folder (`bases/<key>/etat.sqlite`, one per output folder, the key derived from the path), and says so in the log and the status bar: « Folder on a network share: local working database, copied back to the share every 30 s ».

- *At the start:* the database on the share is copied (plain file copy) into the working database.

- *During processing*, every 30 s, *at the end*, on *stop* and *cancel*: the working database is copied back to the share — consistent snapshot, `PRAGMA integrity_check`, copy into a temporary file in the same folder, read back and SHA-256 compared, then replaced in one go (`os.replace`). No lock is ever taken on the share: no orphan lock is possible after a crash, and a reader (another computer, the NAS) always sees a complete database.
- *At the next start*: if the database on the share is newer (processed elsewhere), it is copied in; if the working database holds writes that were not copied back (crash, power cut), they are copied at once.
- *Two writers* (two computers, or the NAS, processed into the same folder meanwhile): nothing is overwritten. Coupole says so clearly and offers a **merge**: for each image the most advanced status is kept (converted > duplicate left out > failed > in progress), the most recent on a tie; pixel fingerprints are pooled. Both original databases are kept next to the working database (`etat.sqlite.local-<date>`, `etat.sqlite.partage-<date>`). Command line: `coupole ohp merge -dest FOLDER`.

Ownership, new images and anomalies read the working database when it is up to date (faster than reading page by page over the network), otherwise the database on the share, read-only. `JOURNAL.txt`, the CSV files and the `LOT.txt` files are still written directly on the share (text files, replaced in one go). The Quality module's cache is only a cache: it stays in the user's cache folder, without copy back (section 8).

Per system. *Linux*: default cifs fails (cause above); mounting with `nobrl` is no longer needed. *macOS*: the smbfs client handles locks differently (depending on how SQLite was built); this could not be checked here, so Coupole does not rely on it and uses the same working database. *Windows*: SQLite can in principle write on a share (native SMB locks), but several writers risk damaging the database; same working database, checked in continuous integration on `\\localhost\C$`. If the database on the share is open in another program when it is replaced, Coupole retries, then waits for the next pass.

4.6 New images in the bank

At startup (at most once a day, adjustable, can be disabled in Preferences), when the Internet is available and a local copy exists in the output folder, Coupole compares a fresh TAP inventory with the copy. If there are new images, a banner appears in the OHP bank (and a message in the status bar): « N new images (M objects, X GB) since DD/MM — Download now? / Later / Show ». *Download* processes only the new images into the same tree; *Show* filters the catalogue; nothing is ever downloaded without explicit consent. The *Check for new images* button runs the comparison on demand. Command line: `coupole ohp new (list)` and `coupole ohp new -download`.

4.7 Astrometric solution

Consistency checks, always made: solution present and readable; scale within 2 % of the nominal one (XPIXSZ/FOCALLEN) or of the night median; angle within 5° of the night median (modulo 180°); centre within 1.5° of the object medoid (fixed) or 2° of the object-and-night medoid (moving). Medians and medoids come from the whole inventory.

With ASTAP (optional), each image is also solved independently:

Status	Meaning
confirmed	ASTAP finds the same centre within 10'' and the same scale within 1 %.
validated	solution kept: checks pass, ASTAP silent (poor or crowded field, short exposure) or absent.
redone	missing or inconsistent solution replaced by ASTAP's (with SIP); the old one goes to HISTORY.
doubtful	checks fail and no ASTAP to decide: image stored apart.
failed	nothing usable: image stored in <code>_sans_solution_astrometrique</code> (<code>_no_astrometric_solution</code> with English names).

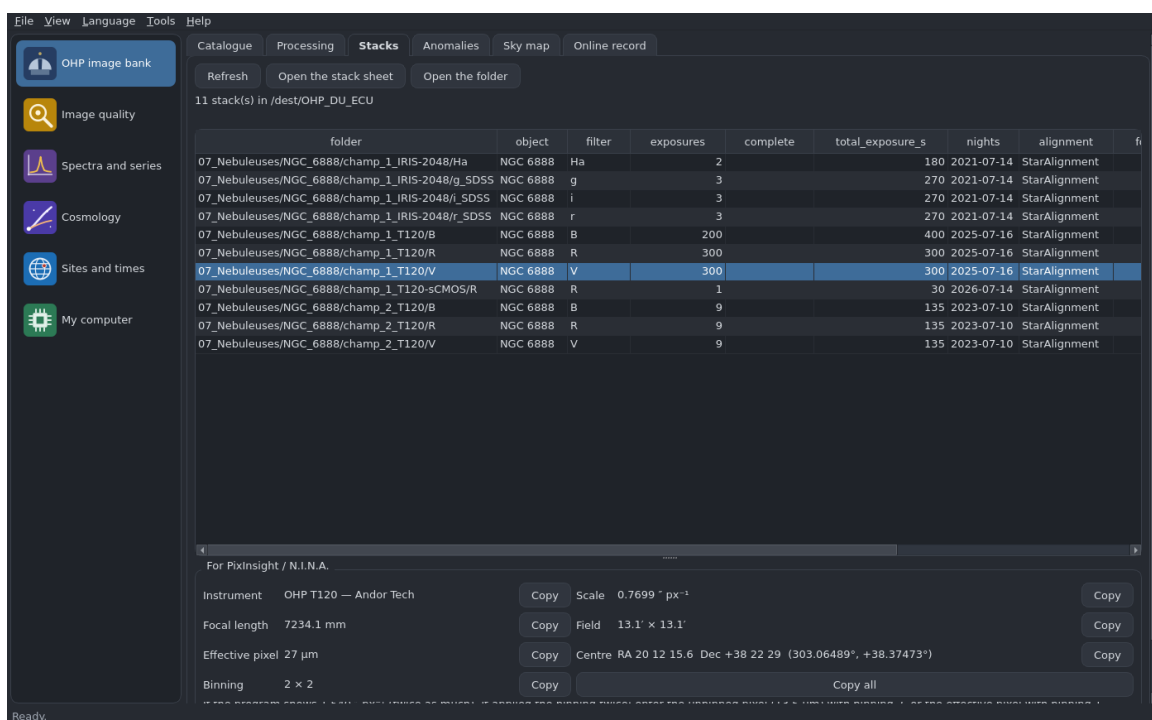
4.8 Fixed headers

All original cards are kept in order; every replaced value is copied into a `HISTORY original value: ... card` (French wording when names are in French).

Keyword	Fix
OBJECT	canonical name (database aliases merged).
FILTER, FILTSYS, FILTERIG	normalised filter (B, V, R, Ha, OIII, g, r, i, z), system (Johnson, Cousins, SDSS, Gunn, Narrowband), original value.
LATITUDE, LONGITUD	fixed only when the swap is proven (swapped values matching the site); correct: untouched; ambiguous: flagged, untouched. Idempotent.
SITELAT, SITELONG, SITEELEV	added when missing (MPC 511 site converted to WGS 84); present and consistent: kept; inconsistent: flagged.
RA, DEC, OBJCTRA, OBJCTDEC	true centre of the solution.
PIXSCALE, ORIENTAT	measured scale, position angle of the Y axis.
FOCALLEN, XPIXSZ	recomputed only when missing or inconsistent by more than 2 %.
CTYPE1/2	« -SIP » without coefficients (IRIS, PinPoint) brought back to TAN.
DATE-OBS, DATEDOUT	added when missing; DATEDOUT = T when the date is shared with another exposure.
XISFCONV	description of the conversion.

The `t_min` column of the database equals `DATE-OBS - exposure/2`; `DATE-OBS` (exposure start) prevails. A header date written in local time (whole-hour offset equal to the site time-zone offset against MJD-OBS/JD, or a « night » exposure dated in daylight) is flagged in the log, never silently fixed.

4.9 Stacks



A stack gathers what is stacked together: same object, same instrument (T120, T120-bin1, T120-sCMOS, IRIS-2048, IRIS-4096), same filter, and same field for a fixed object (all nights) or same night for a moving object.

>_ Command

```
07_Nebulae/NGC_6888/field_1_T120/V/20250716-214427_NGC_6888_V_1s.xisf
01_Asteroids/914_Palisana/night_2023-08-15_T120/R/LOT.txt
```

`LOT.txt` (French then English) gives object, database names, instrument, filter, exposures and total exposure, nights, mean centre, angle, solution status, doubtful dates and stacking advice (StarAlignment

or CometAlignment, rejection suited to the number of exposures). File names: UTC date and time of the exposure start, object, filter, exposure; sorting by name = sorting in time. Names are safe under Windows. The *Stacks* tab adds a *complete* column: converted exposures against the bank's exposures for the same key (object, instrument, filter, and night for a moving object); a fixed object split into several fields counts its stacks together.

Owned $\neq$ stackable together. « 840 / 840 » in the *owned* column says all the images of an object are there, not that they form one stack: the *stacks* column of the object list and its tooltip give the breakdown (« 840 images sorted into 9 stacks: field 1 T120: B 200 · V 300 · R 300; ... — stack each stack separately »), and right-clicking an object offers *Show this object's stacks* (Stacks tab filtered, *All stacks* button) and *Open the target folder* (double-click: the same).

For PixInsight / N.I.N.A. Below the stack table, a box gives, for the selected stack, what to enter in PixInsight (ImageSolver, WBPP *Astrometric solution* panel) or N.I.N.A.: instrument, **focal length** (the one giving the measured scale: $f = 206.265 \times \text{XPIXSZ}/\text{PIXSCALE}$, for instance 7,234.1 mm for the T120 instead of the original header's 7,200), **effective pixel** (binning included: T120 iKon-L in binning 2: 27 μm), binning, **scale** (" px^{-1} "), field and centre, each with *Copy*, plus *Copy all* (also on a stack's right-click: *Copy the astrometry parameters*). The values are read from the header of the stack's images, never hard-coded; the *focal length*, *pixel* and *scale* columns can be hidden by right-clicking the header. *LOT.txt* gives the same values. If a program shows twice the scale, it applied the binning twice: enter the unbinned pixel (13.5 μm) with binning 2, or the effective pixel with binning 1.

Checked in PixInsight's code (PJSR: *AstrometricMetadata.js*, *BPP-Solver.js*): ImageSolver reads the `Instrument:Telescope:FocalLength` (m) and `Instrument:Sensor:XPixelSize` (μm) properties, otherwise `FOCALLEN` and `XPIXSZ`, without multiplying by `XBINNING`; WBPP solves the integrated *master* with its metadata if they survive integration, otherwise with the values of its panel, which stay those of the last instrument used: enter focal length and pixel there (one WBPP settings file per instrument avoids typing them again). Coupole therefore writes `FOCALLEN` matched to the measured scale (old value in `HISTORY`) and the `Instrument:Sensor:XPixelSize/YPixelSize`, `Instrument:Camera:XBinning/YBinning`, `Instrument:Camera:Name`, `Instrument:Telescope:FocalLength` and `Aperture` (T120: 1.2 m) properties. For files already converted: `coupole ohp metadata FOLDER -rewrite` (without `-rewrite`: dry run) completes the headers *without touching the pixels* (block copied byte for byte, read back and compared; atomic writing; log `_traitement/metadonnees.csv`) and notes focal length and pixel in the state database; `coupole ohp sort` then rewrites the *LOT.txt* files. To check a master: open it, *FITSHeader* or *Image > Properties*, look for `XPIXSZ`, `FOCALLEN` and `Instrument:*`.

4.10 Opening an image

Double-click an owned image (image list, Quality table): the system's associated application. Right-click: *Open*, *Open with* and *Open file location* (file selected: Explorer /select, Finder open -R, freedesktop file manager through D-Bus FileManager1.ShowItems — Dolphin, Nautilus... —, otherwise the folder). *Open with* only offers the astronomy programs found on the computer (PixInsight, Siril, ASTAP, Aladin, N.I.N.A.: PATH, usual locations, flatpak, /Applications, Program Files) and able to read this file according to the checked table of section 7; the others are greyed with the reason (« ASTAP only reads uncompressed XISF: choose the FITS format in Processing »). An image not downloaded yet: the status bar says so. A disabled entry carries its reason in its label (« Open with (nothing downloaded) ») and the full reason shows in the status bar on hover: no more tooltip over the menu (on KDE it grabbed the mouse).

How each program is started (0.1.12). Always the detected program, with a *list* of arguments (never a shell); each launch is recorded in `coupole.log` (method, program, arguments, process id; if the program stops within the first 3 seconds: exit code and the beginning of its output) and its result shows in the status bar: « Opening in Siril... », or the error (program not found, immediate stop with its message). Never silent.

- **PixInsight** — Linux: the `bin/PixInsight.sh` script (the publisher states that the binary alone does not start on Linux). This script rebuilds its arguments through an `eval` that only quotes those containing a space: a path with `$`, `'`, `'`, `"`, `\`, `(`, `*`... would be interpreted; Coupole then passes a *temporary symbolic link* with a safe name (to save: « Save As » into the original folder). Windows: `bin\PixInsight.exe`. macOS: `open -a PixInsight.app file`, or `open -n -a PixInsight.app -args -n file` for a new instance.
- **PixInsight already open.** Without options, PixInsight hands the file to the running instance (« Yielded execution to running application instance #1 », code 0); if that instance is busy (script running), *nothing opens*. Hence the setting *Preferences > PixInsight already open*: « Open a new window (-n) » (default, safe) or « Send to the open window (try) ». When an instance is running (process detected), the menu offers both: « PixInsight (new window) » and « PixInsight (open window, try) », the setting's one first; if the output says PixInsight yielded, the status bar says so.
- **Siril**: `siril file`; as a flatpak: `flatpak run -file-forwarding org.free_astro.siril @@ file @@` (the document portal only forwards the file if the sandbox cannot see it). The Flathub package sees the whole system (`-filesystem=host`) except `/tmp`, `/run` (apart from `/run/media`), `/usr`...; if a flatpak *override* removes a folder, the status bar says so and gives the command: `flatpak override -user -filesystem=/mnt/partage org.free_astro.siril`.
- **ASTAP, Aladin**: *program file*.
- **N.I.N.A.** does not open an image given as an argument (its only options: profile, sequence, debug, graphics acceleration): it stays greyed in *Open with*, with the reason; open the image from within N.I.N.A.

Double-click opens PixInsight without the image (Linux). `xdg-open` starts the application associated with the file type through its `.desktop` file. If its `Exec=` line has no `%F` code (`%f`, `%u`, `%U`), the application starts *without* the file — the case of a hand-written `pixinsight.desktop`. Coupole detects it (`xdg-mime query filetype`, then `xdg-mime query default`, reading the `.desktop`) and then starts the program directly (status bar: « PixInsight started directly... »); another application in that case is not started, a message says why. To fix the whole desktop (Coupole does not change this file), in `~/.local/share/applications/pixinsight.desktop`:

```
Exec=/opt/PixInsight/bin/PixInsight.sh %F
MimeType=image/x-xisf;image/fits;
```

then `update-desktop-database ~/.local/share/applications` and, if needed, `xdg-mime default pixinsight.desktop image/x-xisf`.

Copy from an older processing run. An output made before Coupole (script `ohp_xisf.py`) or on another computer records the absolute path of the time in its state database (« `/srv/ancien/OHP_DU_ECU/...` »). Coupole finds each file in the current output folder, in this order: relative path recorded in the database, `_traitement/journal.csv` (destination column), former root or type folder, then the expected name in the stack folder; paths found through the journal are recorded once and for all in the database (in the background, at first load; through the local working database on a share). Open, *Open with*, file location, target folder and stacks therefore work on a copy of the whole bank (7,625 images found in 0.34 s, recorded in the database in 0.8 s). `coupole ohp metadata FOLDER -rewrite` records these locations too.

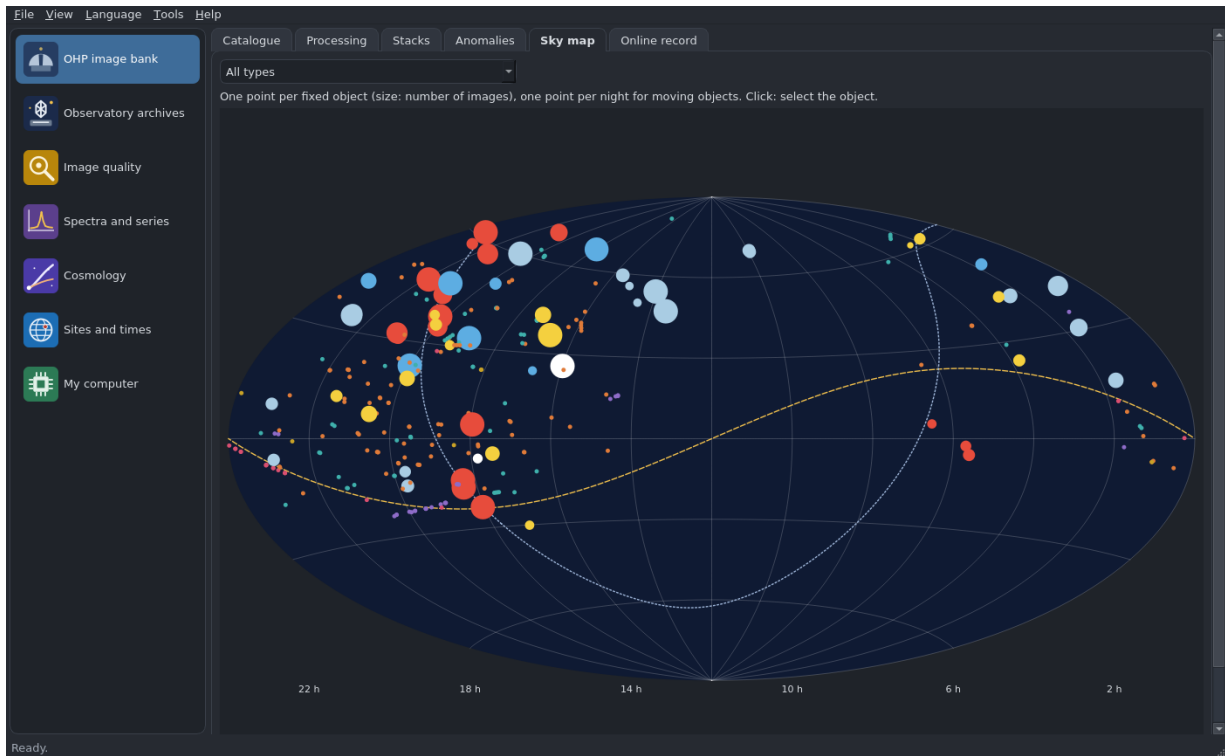
4.11 Anomalies

The screenshot shows the 'Anomalies' tab in the Coupole application. The sidebar on the left contains icons for different data sources: OHP image bank, Observatory archives, Image quality, Spectra and series, Cosmology, Sites and times, and My computer. The main panel displays a table of anomalies. The table has columns for 'anomaly', 'action', 'object', 'night', 'file', and 'detail'. The 'anomaly' column lists various types of anomalies, including 'date_partagee', 'date_diurne', 'champ_incoherent', 'classement', and 'instrument'. The 'action' column shows the status of each anomaly, such as 'flagged' or 'left out'. The 'object' column lists the astronomical objects being observed, such as 'C/2021 S3 (PANSTARRS)', 'IC 5070', 'NGC 5981', and 'RS Ophiuchi'. The 'night' column shows the date and time of the observation. The 'file' column shows the full path to the image file. The 'detail' column shows a small icon representing the anomaly. At the bottom of the window, a status bar indicates 'Ready'.

Kind	Action	Meaning
meme_fichier	left out	same exposure under another path or spelling
copie_diurne	left out	daytime-dated copy of a correctly dated image
pixels_identiques	left out	same pixels (SHA-1) as another image, at processing
memes_meta	flagged	same date, filter, exposure, size, different name
date_partagee	flagged	DATE-OBS identical to another exposure
date_diurne	flagged	Sun above the horizon at the site (never for the Sun or radio data)
champ_incoherent	flagged	centre far from the other exposures of the object
classement	flagged	name classified automatically or not at all
instrument	flagged	instrument missing from the table

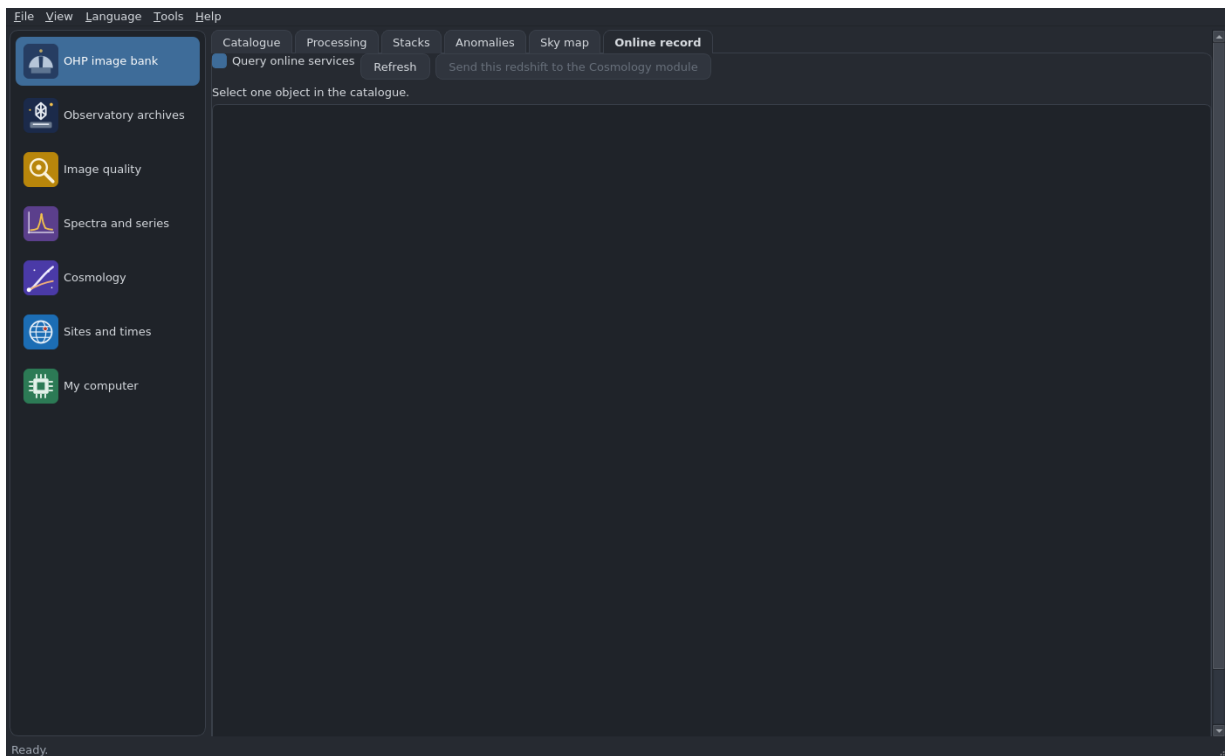
On 8 October 2026: $133 + 11 = 144$ duplicates left out, 1,654 shared dates, 3 daytime exposures, 92 inconsistent fields. CSV export with bilingual columns; nothing is deleted.

4.12 Sky map



Aitoff projection, right ascension increasing to the left, 12 h at the centre; one point per fixed object (size: number of images) and per night for moving objects; ecliptic dashed, galactic plane dotted.

4.13 Online record (optional)



When the Internet is available, the *Online record* tab (or the button of the same name under the catalogue) shows public information on the selected object:

- deep sky and stars: **SIMBAD** (CDS, Strasbourg) — type, ICRS coordinates, magnitudes, parallax (and the distance it gives when precise enough), preferred measured distance with its

method and reference, radial velocity and redshift, angular size, spectral and morphological types, other names; links to the SIMBAD page, **Aladin Lite** (sky image centred on the object) and, for galaxies, **NED**; when SIMBAD does not know the name, **Sesame** (CDS) also queries NED and Vizier;

- asteroids, comets, trans-Neptunian objects: **JPL** Small-Body Database — orbit class, near-Earth or not, elements (a , e , i , q , Q , Ω , ω , period, epoch), diameter, albedo, rotation period, absolute magnitude;
- planets and NEOCP designations: no record (explanatory message).

The identifier chosen by the service is always shown: SIMBAD, for instance, attaches « NGC 6888 » to its Wolf-Rayet star HD 192163; the record says so. A positive redshift can be sent to the Cosmology module (*Send this redshift to the Cosmology module* button).

Politeness and offline use. Nothing is sent while the tab is hidden; one request per selected object (after 0.6 s without another selection change), at most one per second and per service, 8 s timeout, Coupole's User-Agent. Answers are kept for 30 days in Coupole's cache (`enligne.json`, with their date; « not found » for one day). Offline, the last known record is shown with its date, otherwise a clear message; the rest of the application never depends on it. The *Query online services* box stops all requests. The addresses (`simbad.base`, `simbad.page`, `aladin.page`, `ned.page`, `sbdb.page`, `resolution.sbdb`, `resolution.sesame`) are set like the other sources.

5.1 Searching

A name (resolved by SIMBAD then Sesame), coordinates in degrees (83.82 -5.39) or sexagesimal (05:35:17 -05:23:28), a radius (3 arcminutes by default) and the archives to query (MAST, ESO and IRSA by default). For Voyager, Cassini and Juno: a Solar System body (Jupiter, Saturn, Io, Enceladus...) with their archives ticked. Filters: instrument (part of the name: NIRCAM, ACS), filter (exact name: F444W), observation dates. Two boxes ticked by default: *Final products only* (JWST _i2d, Hubble _drz/_drc, Phase 3, SEIP, Atlas) and *Public data only* (release date in the past). An observation is kept if its centre lies in the circle or its footprint contains the position; closest first. The archives answer in parallel; one that is unreachable, requires an account or has no interface is reported in the status bar without stopping the others. An answer truncated by the limit (2,000 rows per archive on the command line, 20,000 in the interface) is reported: refine the search.

Record. Clicking a row shows the thumbnail provided by the archive (cached) and the record: mission, instrument, filter, wavelength, observation and release dates, target, programme, principal investigator, title, calibration level, file, size, **credit** and **usage terms**, link to the archive page. Badges: green = already downloaded and prepared, orange = failure, arrow = to download; greyed row: not yet public.

5.2 Downloading and preparing

Select rows, then *Estimate the volume*: announced sizes (ESO, IRSA, NOIRLab, PDS) are added up, the others measured by a one-byte request (beyond 60 files, a sample per instrument, extrapolated and marked « $\approx$ »). Real orders of magnitude: a NIRC*Cam i2d* of the Pillars of Creation 5.4 GB, a MIRI *i2d* 153 MB, a WFC3/IR *drz* 37 to 87 MB. *Download the selection* estimates first and asks for confirmation beyond 2 GB (archives_seuil_go setting). Capped rate (8 MB/s, archives_debit_mo_s), two files at a time, at most one search request per second and per service; an interrupted file resumes (partial request), except on a server refusing them (ESO, KOA) where it restarts. *Pause* and *Cancel* are immediate.

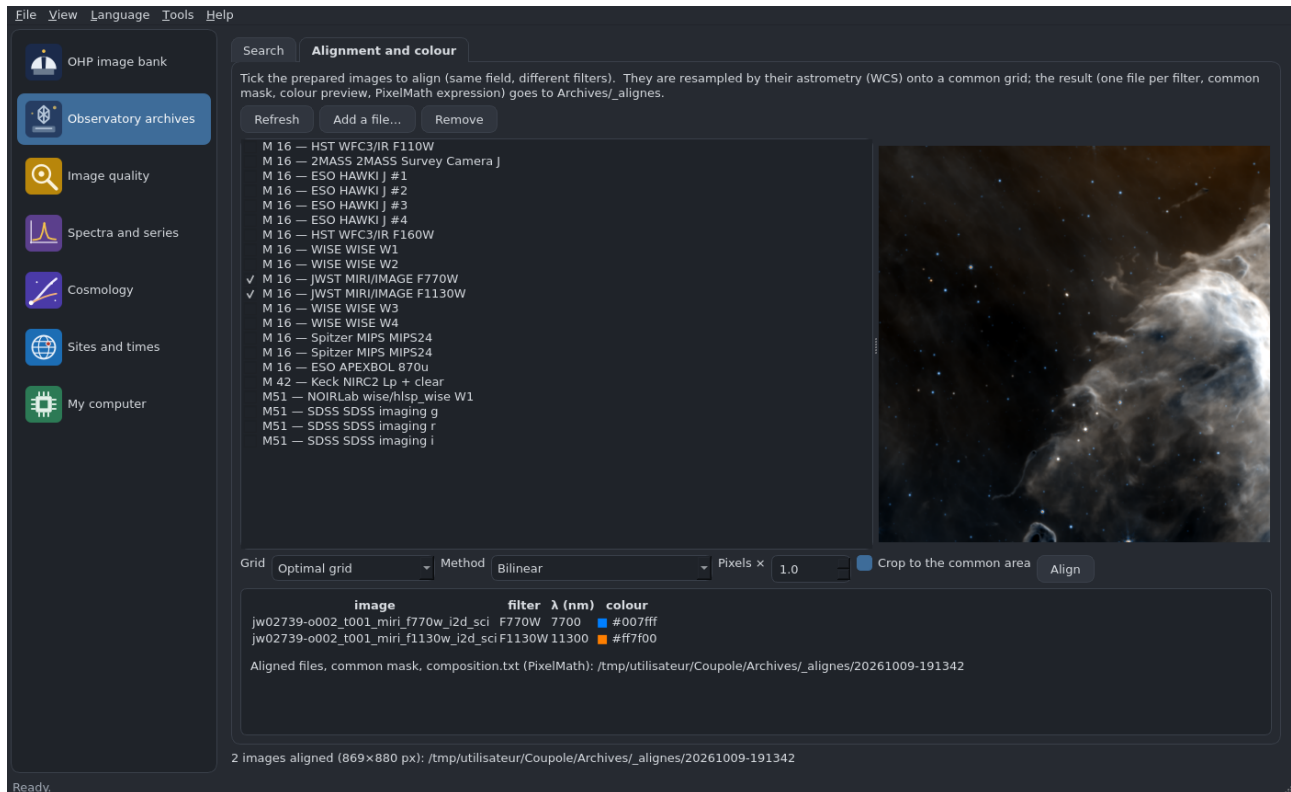
Each file is filed in <folder>/Archives/<mission>/<target>/<instrument>/<filter>/ and prepared:

- science extension extracted (SCI; otherwise the first image), one file per detector for an ESO multi-detector exposure (_d1, _d2...);
- header: WCS (SIP distortions included), BUNIT, photometric constants (PHOTMJSR, PIXAR_SR, PHOTFLAM...), telescope, instrument, filter, dates, exposure time, programme, investigator, then ARCHIVE, ARCHID, ORIGFILE, LAMBDA, CREDIT and the usage terms as a comment;
- pixels without data (NaN: mosaic edges) set to 0, mask _masque.fits (1 = data);
- XISF Float32 (bounds = data minimum and maximum: physical values are kept) or FITS Float32;
- original file kept (unticked: deleted once the image is prepared).

The state is kept in Archives/_etat/etat.sqlite (same network-share logic as the OHP bank, section 4.5) and the bilingual log in Archives/_etat/JOURNAL.txt; a prepared file is never redone. Double-click: open the prepared image; right-click: *Open with*, location, archive page.

Planetary formats. Voyager and Cassini products are VICAR files with a PDS3 label; Juno products are PDS3 files; PDS4 (XML label) is read too. Coupole reads these formats itself (8 to 32-bit integers, IEEE reals, byte order, line prefixes, scaling), flips the rows so the image shows as in the archive and writes FITS or XISF. Peculiarity: the PDS3 label of the calibrated Cassini images places the image one line too early (a VICAR binary line precedes it); the data are read from the VICAR header. Not handled (clear message): compressed files (Voyager EDR), VAX reals, multi-band BIL/BIP interleaving, cubes; a JunoCam image is converted as is (strips of 128 lines per filter), without geometric reconstruction.

5.3 Aligning and colouring



Archive images have an exact astrometry: no star matching (stars change from one instrument and wavelength to another); each image is **resampled** onto a common grid: the grid of one image (often the sharpest filter) or an *optimal grid* covering them all, at the finest pixel scale, optionally enlarged (*Pixels* ×). Methods: bilinear (default, fast), adaptive (anti-aliased, better when the pixel scales differ a lot), exact (exact pixel overlap, slow; below 0.05'' per pixel, adaptive replaces it), nearest neighbour. Adaptive and exact need the optional **reproject** library (`pip install "[alignement]"`); without it, bilinear is done by `scipy`. A **per-pixel** quantity (Hubble electrons/s, SDSS nanomaggies) is multiplied by the pixel-area ratio; a surface brightness (JWST MJy/sr) is kept as is. Optional crop to the largest rectangle covered by every image (a dead column in the middle of a mosaic does not shrink it). The accuracy is that of the archives' astrometry: a residual offset of about a pixel may remain between missions; star alignment in PixInsight or Siril on the resampled files then only has a translation left to correct.

Output in `Archives/_alignes/<date>/`: one file per filter (`_aligne`), the common mask, a colour preview (`aperçu_couleur.png`, asinh stretch) and `composition.txt`. Filters are sorted by wavelength (*chromatic ordering* of the Hubble and JWST press images): three filters give blue, green and red; more give hues spread from blue to red. `composition.txt` gives the colour table and the PixelMath expression (image identifiers as PixInsight creates them when opening); in Siril, the RGB composition uses the same, already aligned files.

5.4 Command line

```
coupole --lang en archives list
coupole --lang en archives search "M 16" --radius 6 --mission JWST --instrument MIRI
coupole --lang en archives estimate "M 16" --mission JWST
coupole --lang en archives download "M 16" --mission JWST --instrument MIRI --dest ~/Sky
coupole --lang en archives download M51 --archive sdss --filter g --filter r --filter i --dest
coupole --lang en archives search --archive opus --body Io --from 1979-03-01 --to 1979-03-10
coupole --lang en archives prepare file_i2d.fits --format fits
coupole --lang en archives align A_sci.xisf B_sci.xisf C_sci.xisf --dest ~/Sky/Pillars --optim
coupole --lang en archives status --dest ~/Sky
```

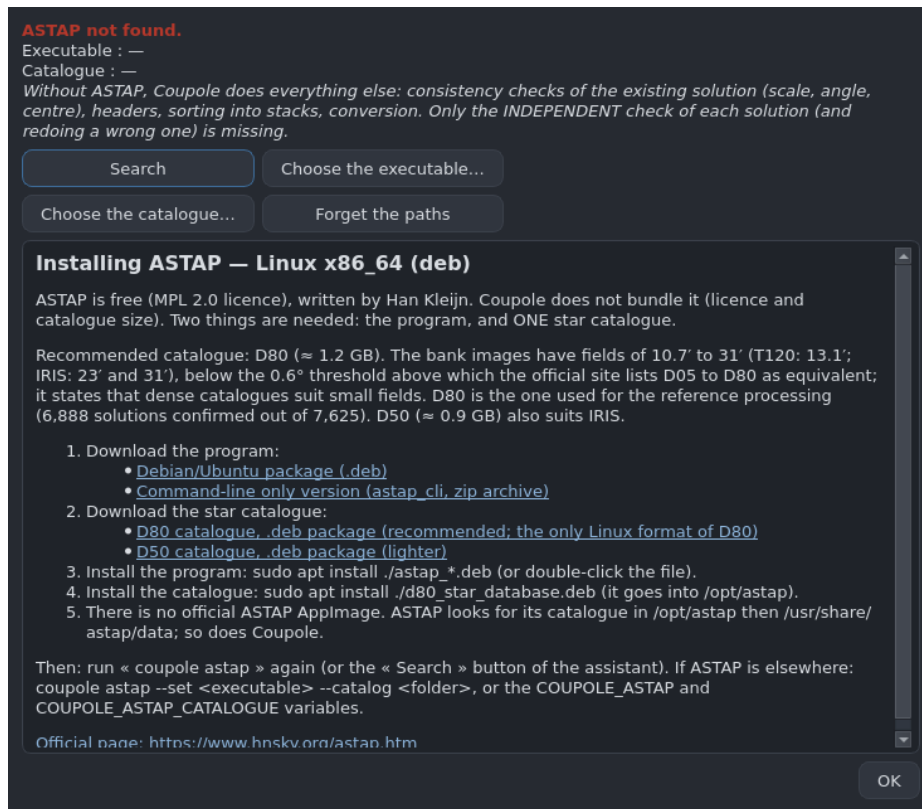
`-id` keeps precise observations (identifier, with or without the archive prefix, or beginning of the file name); `-all-levels` also shows calibrated and raw frames; `-json` and `-csv` for scripts. Beyond the threshold, `download` needs `-yes`.

Credits. Any image published from these data must cite the archive and the mission: « NASA/ESA/CSA, STScI » (JWST), « NASA/ESA, STScI » (Hubble), « ESO » and the programme ID, « NASA/JPL-Caltech » (Spitzer), « NASA/JPL-Caltech/UCLA » (WISE), « 2MASS/UMass/IPAC-Caltech/NASA/NSF », NOIRLab, KOA, SDSS, and for the probes « NASA/JPL-Caltech » (Voyager), « NASA/JPL-Caltech/Space Science Institute » (Cassini), « NASA/JPL-Caltech/SwRI/MSSS » (Juno). Coupole writes it in the record and in the header of every file (CREDIT).

CHAPTER 6

ASTAP (optional)

Without ASTAP, Coupole does everything else. ASTAP (Han Kleijn, MPL 2.0 licence, <https://www.hnsky.org/astap.htm>) adds an independent check of each solution. It needs the program and one star catalogue.



Catalogue. The official site states that for a field of 0.6° or more, D05, D20, D50 and D80 all suit, and that dense catalogues serve small fields. The bank's fields range from $10.7'$ (0.18°) to $31.4'$ (0.52°): **D80** is recommended (about 1.2 GB; catalogue of the reference processing, 6,888 solutions confirmed out of 7,625); D50 (about 0.9 GB) also suits IRIS.

System	Program (links of the official page)
Windows x64	astap_setup.exe then d80_star_database.exe (same folder, C:\Program Files\astap); optional: astap_command-line_version_win64.zip
Windows ARM64	astap_command-line_version_win11_aarch64.zip in C:\astap; catalogue: d80_star_database.exe (x86 installer, emulated) into that folder, or d50_star_database.zip (no zipped D80)
macOS Apple Silicon	astap_M1.pkg, d80_star_database.pkg; once: codesign -force -s - /Applications/ASTAP.app/Contents/MacOS/astap
macOS Intel	astap.pkg, d80_star_database.pkg
Linux Debian/Ubuntu	astap_amd64.deb (or astap_aarch64.deb), d80_star_database.deb (into /opt/astap)
Linux Fedora/openSUSE	astap_amd64.rpm; D80 extracted from the .deb (below) or zipped D50 into /opt/astap
Linux Arch/Manjaro	astap_amd64_gtk3.pkg.tar.zst (or GTK2); D80: AUR package d80-star-db-astap or extracted from the .deb; D50: d50_star_database.pkg.tar.zst
Other Linux	astap_amd64.tar.gz or astap_command-line_version_Linux_amd64.zip; no official AppImage

No zipped D80. The « D80 (zipped) » link of the official page is dead (404, checked on 9/10/2026): D80 only exists as `.exe`, `.pkg` and `.deb`. Outside Debian, extract it from the `.deb` without touching the permissions of `/` (command checked on the real package: 1,213,392,756 bytes, 1,476 tiles):

>_ Command

```
bsdtar -xf d80_star_database.deb data.tar.xz
sudo mkdir -p /opt/astap
sudo tar -xJf data.tar.xz -C /opt/astap --strip-components=3 --no-same-owner
-wildcards './opt/astap/d80_*
```

(never `sudo tar -x ... -C /`: the archive contains `./` and `./opt/`). **Arch/Manjaro:** the official package puts `astap` and `astap_cli` in `/opt/astap` only (nothing on the `PATH`: Coupole finds them there); the GTK2 graphical `astap` needs `libgtk-x11-2.0`, missing from the official repositories (AUR only) and not declared; `astap_cli` is self-contained (statically linked) and is all Coupole needs; for ASTAP's interface, the GTK3 package. A continuous-integration test checks every link, for every system.

Detection. Order: chosen path (Preferences, `coupole astap -set`), `COUPOLE_ASTAP` variable, `astap_cli` everywhere (`PATH` then default locations), only then `astap`, default locations (Windows Program Files\astap, %LOCALAPPDATA%\astap; macOS /Applications/ASTAP.app, /usr/local/opt/astap; Linux /opt/astap, /usr/bin, /usr/local/bin, ~/astap). Catalogue: chosen folder, `COUPOLE_ASTAP_CATALOGUE`, program folder, /opt/astap, /usr/share/astap/data, /usr/local/opt/astap. A catalogue is declared complete with its 1,476 tiles; the version is read from `astap_cli -h`.

CHAPTER 7

Output formats

Format	Content	Readers
XISF (default)	Float32 in ADU, <code>bounds="-1000:65535"</code> ; UInt16 for integer IRIS FITS; <code>zstd+sh</code> level 9 compression; <code>Observation:*</code> , <code>Instrument:*</code> properties	PixInsight
Compatible XISF	UInt16 + 1,000 ADU pedestal (<code>PEDESTAL</code>), <code>zlib+sh</code> (what N.I.N.A. writes)	PixInsight, Siril ≥ 1.4 , N.I.N.A.
FITS <code>.fits.fz</code>	lossless tile compression: <code>GZIP_2</code> without quantization (floats), <code>RICE_1</code> (integers)	Siril, astropy, DS9
FITS	uncompressed float32 (or UInt16)	all

XISF: `bounds` sets the range PixInsight maps to $[0,1]$ when opening: nothing to set, same scale for every exposure; the $-1,000$ lower bound keeps the negative values of short calibrated exposures. Size: 36 % of the original FITS for the T120 (float64), 62 % for IRIS. **Compressed FITS**: measured with astropy 8.0, `RICE_1` and `HCOMPRESS_1` quantize floats (one-ADU error, even with `quantize_level=0`); only `GZIP_1` and `GZIP_2` without quantization restore floats bit for bit. **float32 FITS**: in PixInsight, set the FITS input range to $0-65,535$ with `Truncate`. For every format except the compatible XISF, the difference to the original FITS stays below half a float32 ULP (checked pixel by pixel).

Compatible XISF (N.I.N.A., Siril). `UInt16 = ADU + 1,000` rounded, clipped to $[0, 65,535]$; the pedestal keeps the negative backgrounds of short calibrated exposures and is written in the header (`PEDESTAL`, PixInsight's convention). Measured on 66 exposures of the bank (`outils/audit2/mesure_perte_uint16`) rounding ≤ 0.5 ADU, background mean changed by at most 0.014 ADU (background noise: 10 to 109 ADU); pixels clipped high: at most 0.012 % (stars above 64,535 ADU); no pixel lost low any more. Without the pedestal, a 10-s T120 exposure with an over-subtracted background (-220 ADU) lost 99.75 % of its pixels to 0 and 6.1 % of the pixels of 1-2-s exposures fell to 0: hence the pedestal. Integer IRIS images stay lossless.

Who reads what (checked)

File	PixInsight	Siril	N.I.N.A.	ASTAP	Note
XISF Float32 (default)	yes	no	no	no	Siril ≥ 1.4 reads it but outside its $[0, 1]$ range; N.I.N.A.: white image
Compatible XISF (UInt16, <code>zlib</code>)	yes	$\geq 1.4.0$	yes	no	ASTAP: uncompressed XISF only
XISF UInt16 <code>zstd</code> (integer IRIS)	yes	$\geq 1.4.0$	no (3.2)	no	<code>zstd</code> read by N.I.N.A. since January 2026 (development)
FITS Float32	yes	yes	no	yes	N.I.N.A. assumes $[0, 1]$; Aladin, astropy: yes
FITS UInt16	yes	yes	yes	yes	
<code>.fits.fz</code> Float32 (<code>GZIP</code>)	—	yes	—	no	ASTAP: Rice only
<code>.fits.fz</code> UInt16 (Rice)	—	yes	—	$\geq 2026.07.16$	—: not checked

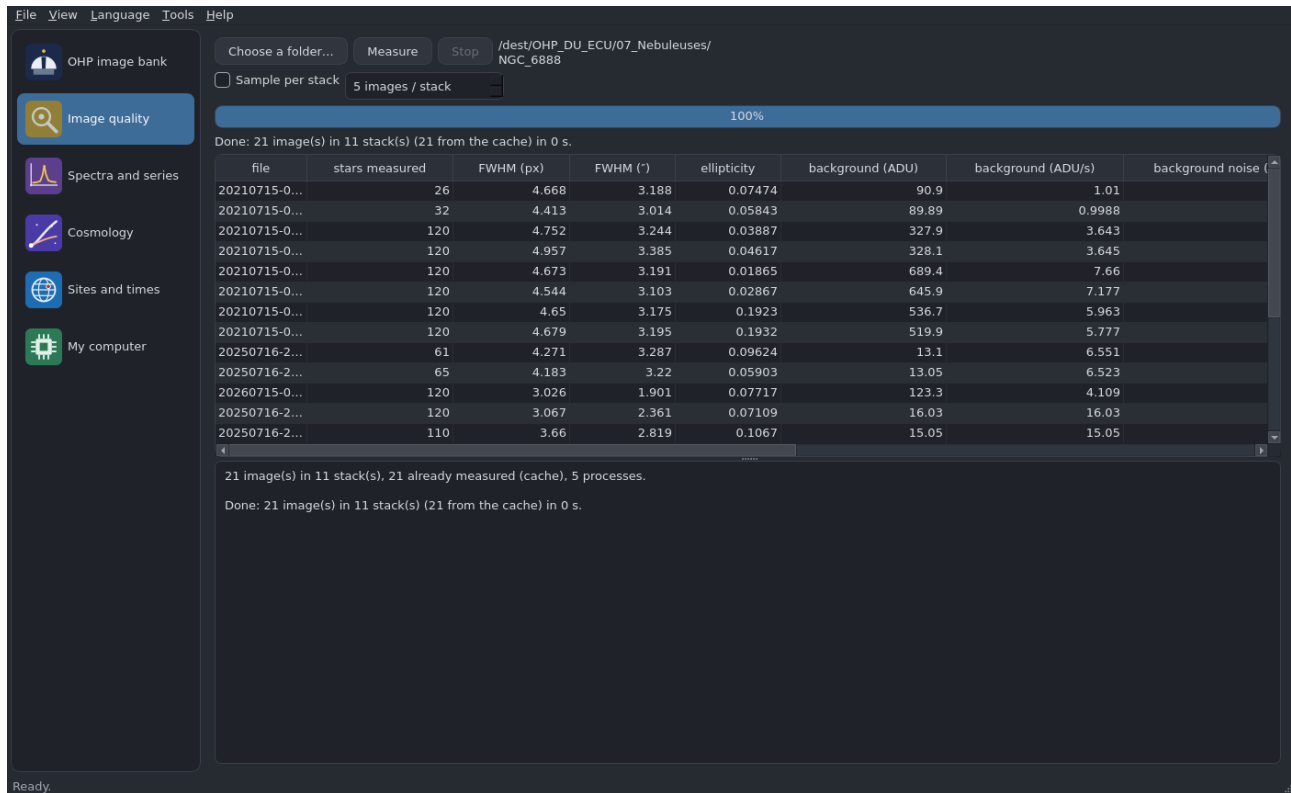
Sources: **PixInsight**, PCL code (`XISF.cpp`: `zlib`, `lz4`, `lz4hc`, `zstd` codecs, with or without `+sh`; `XISFReader.cpp`: `bounds` mandatory for a float, clipped then mapped to $[0, 1]$); XISF 1.0 specification revision 1 (September 2026, <https://pixinsight.com/doc/docs/XISF-1.0-spec/XISF-1.0-spec.html>), XSD schema identical to the one Coupole uses. **Siril**, real test of the official AppImages (siril-cli, pixels read back and compared): 1.2.0 and 1.2.6 do not read XISF (« file type not supported »), 1.4.0 and 1.4.4 read all our codecs bit for bit; but XISF floats stay in ADU there (statistics multiplied by 65,535), whereas float FITS are mapped to $[0, 1]$; Debian 13 and Ubuntu 24.04 packages (Siril 1.2.x): no XISF. **N.I.N.A.**, code (`Version-3.2` tag, 27/11/2025: `XISF.cs` l. 304-328: `lz4`, `lz4hc`, `zlib`, with or

without `+sh`; `Float32Converter.cs` l. 23: `(ushort)(v × 65535)`, `bounds` never read) and real decoding of one of our files. **ASTAP**, official page (« uncompressed » XISF, Rice `.fits.fz` since 2026.07.16, not GZIP). Test files for testers: `tests/donnees/` (synthetic image as default XISF, compatible XISF and FITS, under 300 KB each).

Deviation from revision 1. Coupole writes the centre of the astrometric solution in `Observation:Center:RA/Dec` (choice of the reference processing, older than the revision); revision 1 keeps these properties for the point the telescope was aimed at and standardizes the solution in the `AstrometricSolution:` namespace (not written: the solution stays in the WCS keywords, which PixInsight reads). Kept as is in 0.1.9 so as not to change files compared with the reference processing.

CHAPTER 8

Image quality (optional)



The module needs the SEP library (Source Extractor in Python, LGPL, about 1 MB); it never runs automatically (*Measure* button, *Check image quality* box of the processing, or `coupole qualite FOLDER -write`). Report per stack: `QUALITE.csv` and `QUALITE.txt`.

Sky frames only. By default only sky frames (lights) are measured. Left out without being opened: Flat(s), Dark(s), Bias, Offset(s), DarkFlat(s), Calibration, Master(s) folders (« FLAT L », « Dark-Bias » included), cosmetized, calibrated, registered, debayered, `_platesolve...` and their French names (Plats, Noirs, Biais); `FLAT_`, `DARK_`, `BIAS_`, `DARKFLAT_` files (N.I.N.A., ASIAIR), masters (`masterDark`, `masterFlat`, `masterLight`, `superbias`) and WBPP outputs (`_c`, `_cc`, `_r...`). As a last resort, a frame whose header (IMAGETYP, FRAME or the XISF property `Observation:Image:Type`) says flat, dark, bias, offset or master is left out when measured; Light, Light Frame, Science or Object are measured. The summary says « 70 image(s) found, 66 sky frame(s) measured, 0 with an error, 4 calibration file(s) excluded »; *Excluded files...* lists them with the reason. *Also include calibration frames* box (command line: `-with-calibration`): everything is measured.

Other programs' files, errors. XISF files from N.I.N.A. (keywords without a comment), the ASIAIR (no Metadata), PixInsight and Siril are read by a tolerant reader: only what really prevents reading the pixels is refused; our own files are still checked strictly. A file that cannot be measured shows in the table, *status* column, with its reason (« error: XISF format not read (...) »); the summary counts the errors and `coupole.log` keeps the stack trace; the command line exits with code 1 when no image could be measured. N.I.N.A. frame from a CMOS camera ($9,576 \times 6,388$, 61 Mpx): 4.2 s, 1,170 MB peak (float32 computation), FWHM $3.5''$ at $1.695''/\text{px}$; the number of processes is reduced from the image size (half of the free memory) and the number of fitted stars grows with the sensor (120 to 400).

Speed, resume, sample. Measurements run in separate processes (count given by the machine plan: cores, memory, economy mode and the Preferences limit; at most 3 when the folder is on a network share, detected and announced, since reading then limits the speed), one image being read while the others are computed; the interface stays free, *Stop* is immediate. Each measured image is kept at once in the cache `_traitement/qualite.sqlite` by (path, size, date) and shown; its stack's `QUALITE.csv` (atomic write) is rewritten at most every 5 s and at the end of the stack or on stop: starting again redoes nothing, closing then reopening resumes. Above 200 images, Coupole offers a **sample** of 5 images per stack (spread in time: first, last, middle..., adjustable in the tab and with `-sample N`; measured first, while you choose), enough to judge a stack, or *Measure everything* with the duration estimated from the images already measured; progress shows the time left and the rate (images/min). On a 6-core processor: 1.21 → 0.33 s per T120 image (vectorised Moffat model, analytic derivatives, 120 stars per image), i.e. the whole bank in about ten minutes on 5 processes.

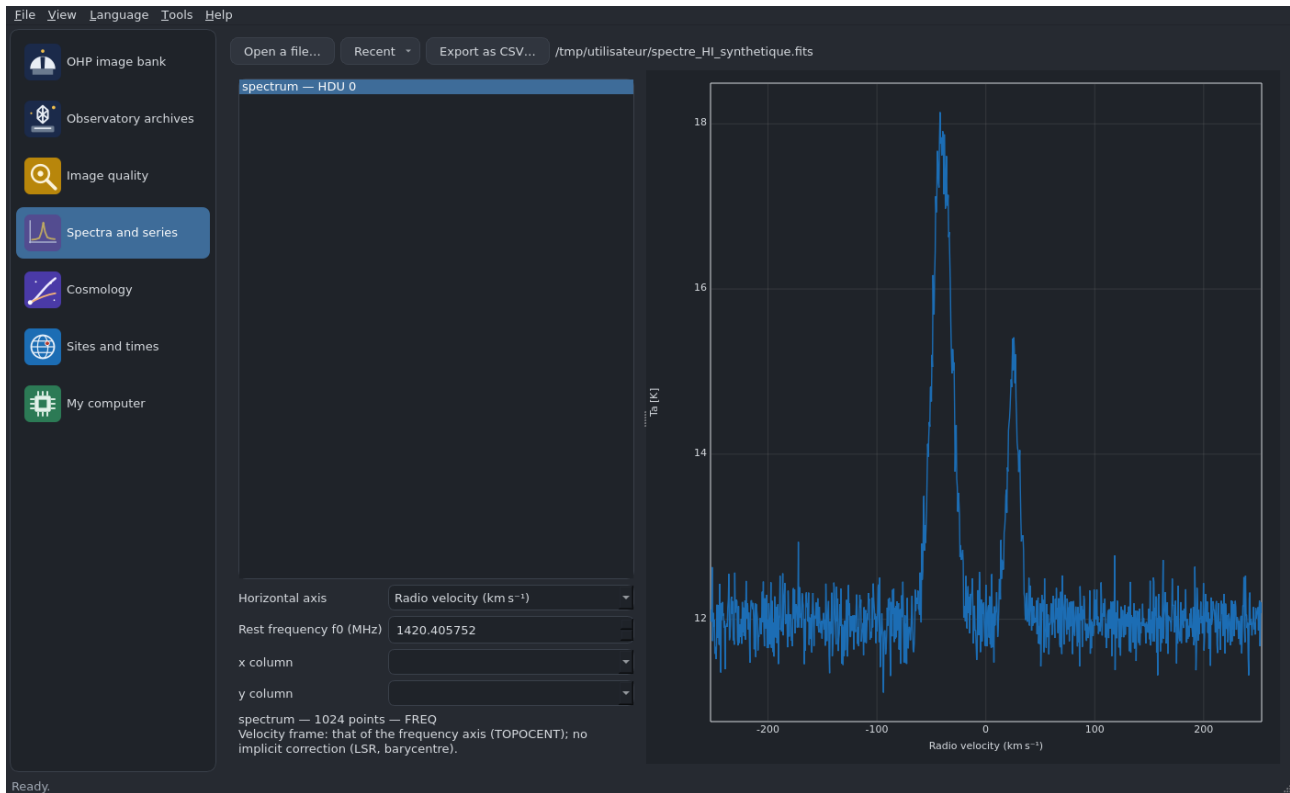
Listing as a stream. Measuring starts with the first stack found, while the listing goes on (« Listing the folder: 1,234 files found in 56 stacks... », *Stop* immediate); the listing opens no file and reads size and date from each folder's listing (at most one **stat** per image). In a Coupole output, only the stack folders are read, from `INDEX_LOTS.csv` or the state database; processing leftovers (**converties**, **telechargements**) are never taken for stacks. On a share, 32 threads read the folders. The cache is **never** written on a network share: Linux's SMB client forwards SQLite's locks to the server, which refuses them (« database is locked » after 30 s, three times before the first measurement in 0.1.8: the 90 s of « Listing the folder... » seen on a real share); it goes to the user's cache folder (an old cache on the share is read). Measured on a real Samba share, 831 XISF of 09_Galaxies: 0.1.8 92.7 s before the first measurement (10 ms latency); 0.1.9 full listing in 0.24–0.57 s (10 ms) and 0.57–1.05 s (20 ms), first result in 0.2–0.6 s.

Measure	Method	Synthetic validation
FWHM	elliptical Moffat with free β fitted to each isolated, unsaturated star (model integrated over the pixel); median	≤ 0.4 % (Gauss and Moffat, 2.5–8 px)
ellipticity	$1 - b/a$ of the same fit	± 0.02
3×3 map	medians per ninth of the field	order recovered
background, noise	SEP, 64 px mesh	0.5 %; 5 %
gradient, residual	plane fitted to the background map; residual amplitude	gradient and vignetting recovered
saturated	peak above SATURATE/DATAMAX (else 60,000 ADU)	5 out of 5
trails	objects with major axis ≥ 12 px and $a/b \geq 6$	1 out of 1, 0 false
SNR	only with the gain (EGAIN/GAIN)	given or absent
sampling	FWHM < 2 px, $2\text{--}5$ px, > 5 px	—

On 14 IRIS images (12 nights, FWHM from 2.4 to 11.3 px), the measured FWHM and the one written by PinPoint in the header correlate ($r = 0.974$), median ratio 1.046 (0.85 to 1.16): the two estimators differ (PinPoint's method is not described). Not offered for lack of validation: mag arcsec⁻² (no zero point), dust-donut detection. A FWHM growing towards an edge may come from the sensor, field curvature, collimation, but also from seeing, tracking or focus: the map alone cannot tell.

CHAPTER 9

Spectra and series



Formats read: FITS (1D image with a linear CRVAL/CDELTA/CRPIX WCS axis, $1 \times 1 \times N$ radio cubes, binary tables) and CSV or text in columns (optional header, unit in parentheses or brackets). Radio velocity: $v = c(1 - f/f_0)$ (FITS radio convention), f_0 read from RESTFRQ, otherwise the HI line at 1,420.405751768 MHz. The frame is that of the frequency axis (SPECSYS); no LSR or barycentric correction is made implicitly. Formats of teaching radio antennas are added with a reader (CONTRIBUTING.md, « Adding a format »).

CHAPTER 10

Sites and times

The screenshot shows the 'Sites and times' section of the Coupole application. The sidebar on the left contains icons for 'OHP image bank', 'Observatory archives', 'Image quality', 'Spectra and series', 'Cosmology', 'Sites and times' (selected), and 'My computer'. The main area displays a table of sites and a map below it.

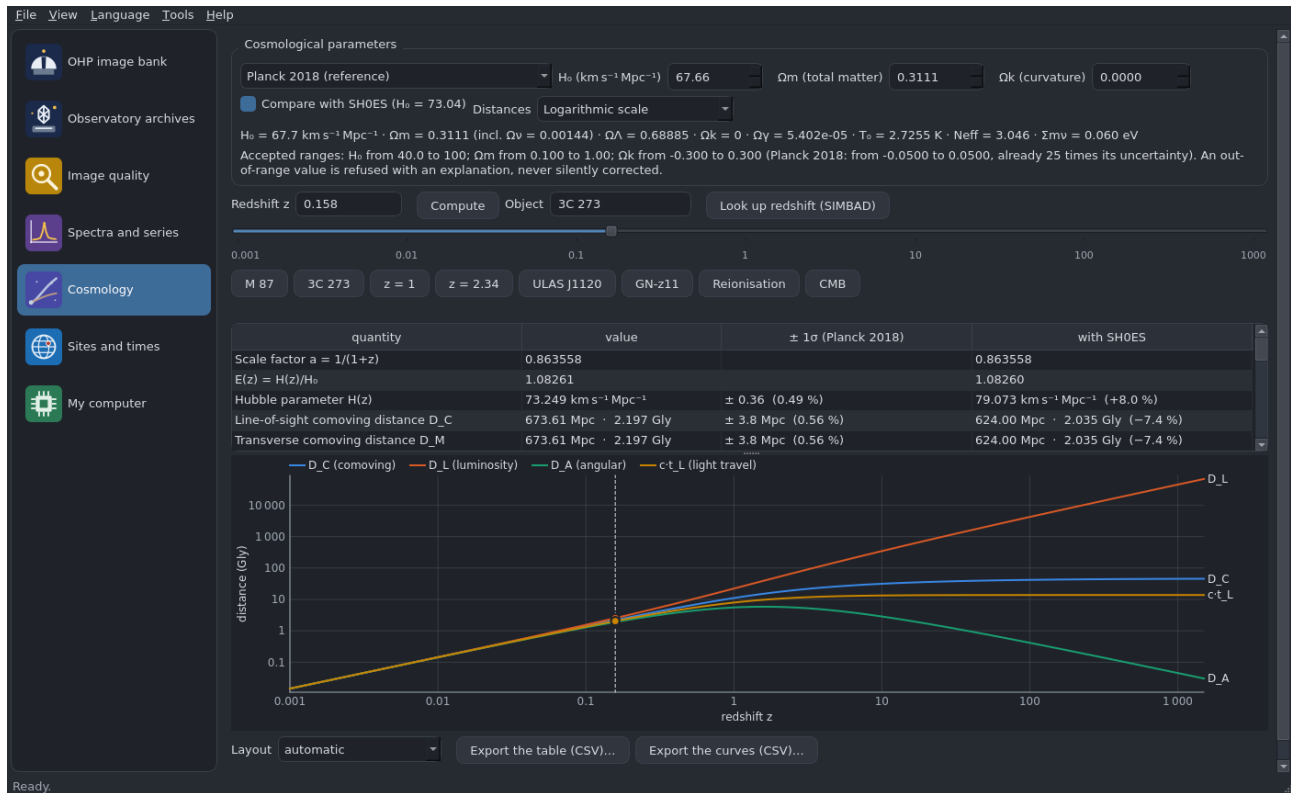
	id	name	latitude (°)	longitude E (°)	altitude (m)	MPC code	time zone	local time	origin
1	ohp	Observatoire de Haute-Provence	43.93187	5.71570	634	511	Europe/Paris	2026-10-09 21:13	shipped
2	meudon	Observatoire de Paris, site de Meudon	48.80507	2.23100	158	005	Europe/Paris	2026-10-09 21:13	shipped
3	paris	Observatoire de Paris	48.83638	2.33675	60	007	Europe/Paris	2026-10-09 21:13	shipped

Below the table is a map showing the locations of the three sites: 'Observatoire de Paris' (red dot), 'Observatoire de Paris, site de Meudon' (red dot), and 'Observatoire de Haute-Provence' (red dot). The map is labeled 'Offline: plain background (map © OpenStreetMap contributors when the network is back)'.

Time is always kept in UTC (aware dates); the site's local time and the user's are shown as well. The « evening date » of a night is the site's local date at the previous noon; daylight is computed from the Sun's altitude at the site (astropy, no IERS download). Shipped sites: OHP (MPC 511), Meudon (005), Paris (007), geodetic coordinates computed from the Minor Planet Center parallax constants. Partner sites are added (IANA time zone, MPC code if any) once their coordinates are known from a reliable source. Map: OpenStreetMap tiles cached on disk, at most two downloads at a time, User-Agent naming the application, attribution shown; offline, plain background and points. On a dense screen (scaling $\geq 150\%$, Mac Retina) the map draws the tiles of the next zoom level at half size (four per logical tile, the « @2x » equivalent): it stays sharp instead of being upscaled by the system.

CHAPTER 11

Cosmology



The redshift z of a distant object does not give « one » distance but several, all correct, answering different questions. The module computes them in the Λ CDM model with

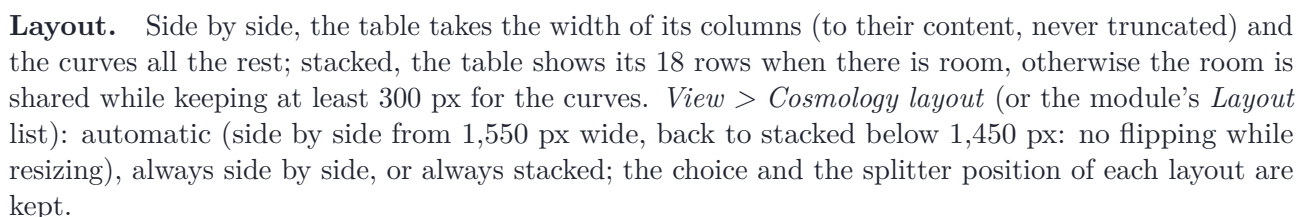
$$E(z)^2 = \Omega_r(z)(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda, \quad D_C = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')},$$

radiation included (CMB photons and neutrinos, one of them massive, 0.06 eV).

Quantity	Use
D_C line-of-sight comoving	present distance of the object; saturates at ≈ 46 Gly (particle horizon)
D_M transverse comoving	equals D_C in a flat universe; enters D_L and D_A
$D_L = (1+z)D_M$	photometry: $F = L/(4\pi D_L^2)$; distance modulus $\mu = 5 \log_{10}(D_L/10 \text{ pc})$
$D_A = D_M/(1+z)$	sizes: $\theta = \ell/D_A$; scale in kpc per arcsecond; maximum near $z \approx 1.6$
$c t_L$	path travelled by the light (the most intuitive, the least physical)
t_L , age at z , t_0	lookback time; check $t_L + t(z) = t_0$
V_C	comoving volume out to z (Hogg 1999, exact in curved universes too)
velocities	cz (naive), relativistic Doppler (inappropriate), $H_0 D_C$ (the only correct one)

Parameters. Planck 2018 by default ($H_0 = 67.66$, $\Omega_m = 0.3111$ incl. $\Omega_\nu = 0.00144$, $T_0 = 2.7255$ K) with the 1σ uncertainty propagated from *correlated* H_0 and Ω_m ($\rho = -0.976$: ignoring it would overestimate the uncertainty) and, optionally, the comparison with SH0ES ($H_0 = 73.04$, Hubble tension). For comparison: Planck 2015, WMAP 9-year, a « textbook » Λ CDM (70 / 0.3 / 0.7 without radiation) and a custom set (H_0 , Ω_m , Ω_k , Planck 2018 radiation). Ranges: H_0 from 40 to 100, Ω_m from 0.1 to 1, Ω_k from -0.3 to 0.3 (± 0.05 for Planck 2018); an out-of-range value is *refused with an explanation*, never silently corrected.

Slider. Under the z field, a slider spans the whole range on a logarithmic scale (0.001 to 1100, marks at 0.01 / 0.1 / 1 / 10 / 100 / 1000). While dragging, the vertical marker of the curves and the table of values follow live: values are then *interpolated* on the curve grid (300 logarithmic points, typical deviation $< 0.5\%$, σ and SH0ES columns suspended) and the status line says so; on release the exact computation resumes. Arrows: fine step (1 % of a decade); Page up/down: one decade. The z field and the slider stay in sync both ways.



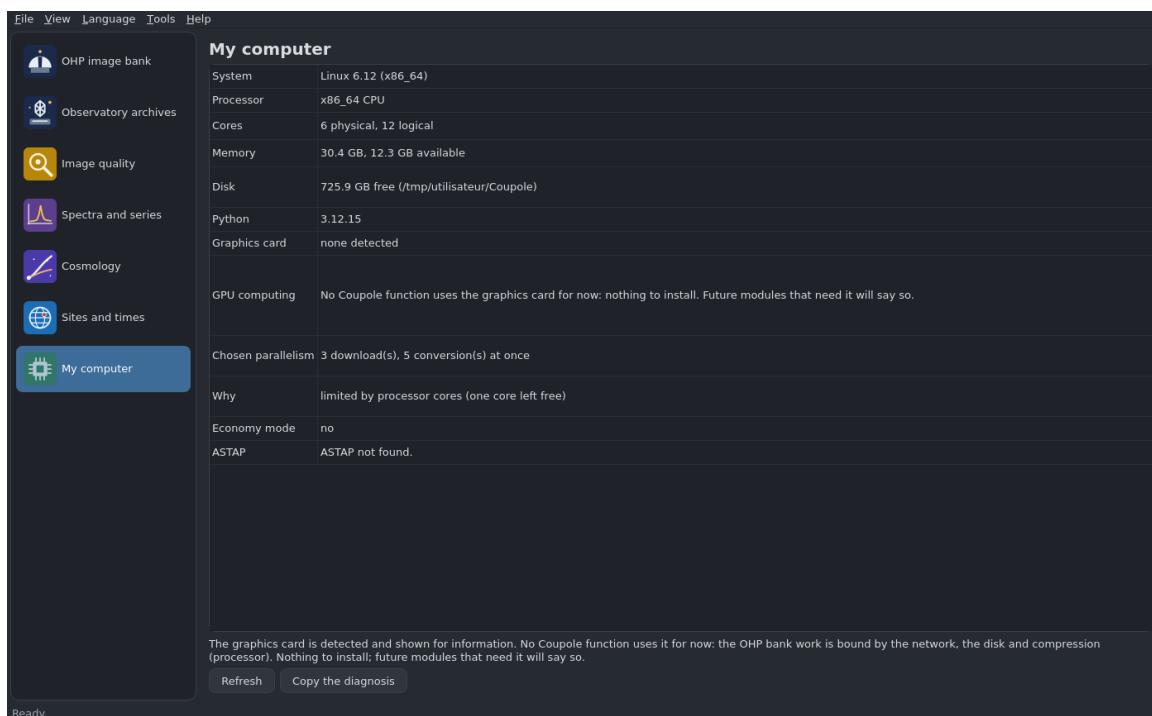
Origin and checks. The computation core is taken from the same author’s « cosmologie-redshift » calculator (public repository <https://github.com/ARP273-ROSE/cosmologie-redshift>), with astropy.cosmology. Before integration it was compared on 8 October 2026 with astropy and with an independent SageMath integration (mpmath, 18 digits, exact Fermi-Dirac function for the neutrinos), from $z = 10^{-8}$ to 1100, for $\Omega_k = 0, \pm 0.01, \pm 0.05$: relative deviation $\leq 3 \times 10^{-6}$ on distances and $\leq 2 \times 10^{-5}$ on ages and $E(z)$ (Komatsu fit of the neutrinos in astropy); no bug. astropy’s comoving volume, however, loses all its digits at very small z in curved universes (down to a negative volume): Coupole uses a series expansion there, matched to the closed form. Scripts: `outils/verif_cosmo/`.

>_ Command

```
coupole cosmo 0.158 -shoes  
coupole cosmo -object "3C 273"  
coupole cosmo 1 2 3 -model wmap9 -csv results.csv  
coupole cosmo -curves 0.001 1500 300 -csv curves.csv
```

CHAPTER 12

Computer and performance



Detection without mandatory dependency: system, architecture, processor, physical and logical cores (bounded by affinity and container quota), total and available memory, free space, graphics cards (nvidia-smi, `/sys/class/drm`, `system_profiler`, CIM). Parallelism: 3 downloads (at most 4), conversions = physical cores – 1 bounded by available memory at 500 MB per conversion (peak measured on an IRIS 4096² image, at most half of the free memory); economy mode (1 conversion, 2 downloads) automatic below 4 GB or 2 cores; manual limits, also bounded by memory. Examples: 2 cores and 2 GB → 2 downloads, 1 conversion; 6 cores and 20 GB free → 3 and 5; 32 cores and 60 GB free → 3 and 16 (cap). The graphics card is not used by the OHP bank (work bound by network, disk and Zstandard); `coupole.core.calcul` offers CuPy to future modules.

Nothing heavy runs on the GUI thread: `astropy`, `scipy` and `SEP` are loaded only when first needed, the machine probes and the ASTAP detection run in the background, and closing the window stops then waits for every thread, task and process (see `docs/AUDIT_2026-10.md`). Large lists stay smooth at 80,000 rows: cells are computed only for the rows shown, sorting uses one key per row (not pairwise comparisons), selecting no longer makes every drawing walk all rows, and changing the theme only recomputes colours (see `docs/AUDIT2_2026-10.md`).

Measurement (6-core processor, NVMe; Debian container)

coupole -version	
window shown (astropy, scipy, SEP not loaded)	0.42 s (1.5 s before)
inventory loaded (background thread)	3.3 s at first launch, 0.1 s after
full chain, 24 T120 images of 8 MB, 8 MB/s cap	24.5 s = duration of the download
same chain without a cap (local server)	2.3 s, 86 MB/s of FITS
conversion of one T120 image (one process)	0.13 s (zstd 14 ms, reading 18 ms, writing and reading 1 ms)
memory peak of a T120 / IRIS 4096 ² conversion (dedicated process)	163 MB / 365 MB (17 MB / 365 MB)
interface during 3 downloads (28 MB/s)	max gap of a 20 ms
<i>Second audit (0.1.4): real bank of 7,625 XISF; network share simulated at 2 ms per access</i>	
final sorting with nothing to move, on the share	1.5 s
48 images to a destination on the share	4.9 s
quality, starting again on the already measured bank (share)	1.9 s
reorganise the bank (reading the 7,625 headers)	57 s
select all, 80,000 images (10 × the bank)	0.2 s (3.1 s before; 0.5 s → 0.17 s)
sort 80,000 images	0.1–0.2 s (3 s before)

CHAPTER 13

Command line

Everything the interface does is available on the command line; `-json` for scripts. Without argument, `coupole` opens the interface. Commands have a French name and an English alias.

>_ Command

```
coupole -lang en ohp estimate "NGC 6888" -night 2025-07-16
coupole -lang en ohp process "NGC 6888" -night 2025-07-16 -filter B -filter V -filter R
-dest ~/OHP -yes
coupole -lang en ohp download Pluto -no-doubtful-dates -dest ~/FITS
coupole -lang en qualite ~/OHP -write
```

coupole

<code>-version</code>	show the version and exit
<code>-lang, -langue {auto,fr,en}</code>	message language (auto: system language)
<code>-v, -verbeux, -verbose</code>	detailed log
<code>-reinitialiser-interface,</code>	erase the saved layout (window, columns, filters, file-dialog folders)
<code>-reset-interface</code>	and exit; settings do not change

coupole astap

Look for ASTAP and its star catalogue, show what to install for this system, or record their location. Coupole works without ASTAP.

<code>-guide</code>	show the installation guide for this system
<code>-definir, -set PATH</code>	path of the ASTAP executable (or its folder)
<code>-catalogue, -catalog FOLDER</code>	star catalogue folder (D80, D50...)
<code>-oublier, -forget</code>	forget recorded paths (back to detection)
<code>-json</code>	JSON output (for scripts)

coupole rapports (reports)

Nothing is sent without consent. Without consent, reports stay in a local folder.

<code>-activer, -enable</code>	allow sending (anonymous) reports
<code>-desactiver, -disable</code>	forbid sending reports

coupole maj (update)

Query the releases published on GitHub. The standalone package updates itself; a pip/pipx installation shows the command to run.

<code>-appliquer, -apply</code>	download and install the update (standalone package)
---------------------------------	--

coupole sources

Without option: list the addresses and their origin (shipped default, remote file, forced value).

<code>-forcer, -set KEY VALUE</code>	force a value (never overwritten by an update)
<code>-default, -default KEY</code>	go back to the default value for this key
<code>-tout-reinitialiser, -reset-all</code>	go back to the default values for every key
<code>-tester, -test KEY</code>	test the connection to this address
<code>-distant, -remote</code>	fetch the sources file published in the repository now

coupole ohp inventaire (inventory)

Bank inventory from the public TAP service (<http://tap-ufe.obspm.fr/tap>, table `ivoa.observ`), kept in a local cache.

<code>-rafraichir, -refresh</code>	query the Observatory's TAP service again (one request)
<code>-manquantes, -missing</code>	list what remains to download compared with the output folder
<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
<code>-json</code>	JSON output (for scripts)

coupole ohp catalogue (catalog)

The 194 database names grouped into objects, sorted by type.

<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-chercher, -search TEXT</code>	keep objects whose name contains this text
<code>-json</code>	JSON output (for scripts)

coupole ohp images (list)

One line per image: date, telescope, filter, exposure, flags (duplicate, shared date, daytime).

<code>OBJECT</code>	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-filtre, -filter F</code>	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
<code>-nuit, -night YYYY-MM-DD</code>	night (evening date); repeatable
<code>-sans-dates-douteuses,</code>	leave out exposures with a shared DATE-OBS or dated in daytime
<code>-no-doubtful-dates</code>	
<code>-tout, -all</code>	the whole bank (≈ 78 GB)
<code>-csv FILE</code>	also write the list to this CSV file
<code>-dest FOLDER</code>	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp estimer (estimate)

Estimate before downloading, with the free space at the destination.

<code>OBJECT</code>	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-filtre, -filter F</code>	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
<code>-nuit, -night YYYY-MM-DD</code>	night (evening date); repeatable
<code>-sans-dates-douteuses,</code>	leave out exposures with a shared DATE-OBS or dated in daytime
<code>-no-doubtful-dates</code>	
<code>-tout, -all</code>	the whole bank (≈ 78 GB)
<code>-format {xisf,xisf16,fz,fits}</code>	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
<code>-dest FOLDER</code>	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp traiter (process)

Full processing of a selection, resumable: running the same command again resumes where it stopped.

OBJECT	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
-type TYPE	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
-telescope {T120,IRIS}	telescope (repeatable)
-filtre, -filter F	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
-nuit, -night YYYY-MM-DD	night (evening date); repeatable
-sans-dates-douteuses,	leave out exposures with a shared DATE-OBS or dated in daytime
-no-doubtful-dates	
-tout, -all	the whole bank (≈ 78 GB)
-dest FOLDER	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
-format {xisf,xisf16,fz,fits}	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
-noms, -names {fr,en}	language of folder names, object names and headers (default: interface language)
-astap {auto,tous,suspectes,jamais,all,suspicious,never}	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
-telechargements, -downloads N	simultaneous downloads (automatic by default; at most 4)
-conversions N	simultaneous conversions (default: from cores and memory)
-econome, -eco	economy mode: 1 conversion, 2 downloads
-debit, -rate MB/s	maximum download rate (default 8 MB/s)
-garder-fits, -keep-fits	also keep the original FITS (in _traitement/fits_origine)
-garder-doublons, -keep-duplicates	also process duplicates (inventory and identical pixels) instead of leaving them out
-oui, -yes	accept a selection over 5 GB without confirmation

coupole ohp tout (all)

The whole bank ($\approx 8,000$ images, ≈ 78 GB of FITS $\rightarrow \approx 30$ GB as XISF), with a volume and time estimate before confirmation; running the same command again resumes where it stopped.

-dest FOLDER	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
-format {xisf,xisf16,fz,fits}	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
-noms, -names {fr,en}	language of folder names, object names and headers (default: interface language)
-astap {auto,tous,suspectes,jamais,all,suspicious,never}	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
-telechargements, -downloads N	simultaneous downloads (automatic by default; at most 4)
-conversions N	simultaneous conversions (default: from cores and memory)
-econome, -eco	economy mode: 1 conversion, 2 downloads
-debit, -rate MB/s	maximum download rate (default 8 MB/s)
-garder-fits, -keep-fits	also keep the original FITS (in _traitement/fits_origine)
-garder-doublons, -keep-duplicates	also process duplicates (inventory and identical pixels) instead of leaving them out
-oui, -yes	accept a selection over 5 GB without confirmation

coupole ohp nouveautes (new)

Query the TAP service again, compare with the local copy (output folder) and list what is new; `-download` processes them into the same tree (never without this option).

<code>-telecharger, -download</code>	download and convert the listed new images
<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
<code>-format {xisf,xisf16,fz,fits}</code>	output format: <code>xisf</code> (PixInsight, default), <code>xisf16</code> (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), <code>fz</code> (lossless compressed FITS), <code>fits</code> (float32)
<code>-noms, -names {fr,en}</code>	language of folder names, object names and headers (default: interface language)
<code>-astap {auto,tous,suspectes,jamais,all,suspicious,never}</code>	ASTAP check: <code>auto/all</code> (every image when ASTAP is present), <code>suspicious</code> , <code>never</code>
<code>-telechargements, -downloads N</code>	simultaneous downloads (automatic by default; at most 4)
<code>-conversions N</code>	simultaneous conversions (default: from cores and memory)
<code>-economie, -eco</code>	economy mode: 1 conversion, 2 downloads
<code>-debit, -rate MB/s</code>	maximum download rate (default 8 MB/s)
<code>-garder-fits, -keep-fits</code>	also keep the original FITS (in <code>_traitement/fits_origine</code>)
<code>-garder-doublons, -keep-duplicates</code>	also process duplicates (inventory and identical pixels) instead of leaving them out
<code>-oui, -yes</code>	accept a selection over 5 GB without confirmation

coupole ohp reorganiser (reorganise, reorganize)

Walk a folder, recognise files produced by Coupole (header) and move them into the stack tree of `-dest`; name conflict → suffix and a log line; nothing is overwritten.

<code>FOLDER</code>	folder holding the files to sort
<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)

coupole ohp telecharger (download)

Layout: `<folder>/<object>/<night>_<telescope>/<original file>`.

<code>OBJECT</code>	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
<code>-type TYPE</code>	category: <code>ast</code> , <code>neocp</code> , <code>com</code> , <code>pla</code> , <code>tno</code> , <code>pn</code> , <code>neb</code> , <code>amas</code> , <code>gal</code> , <code>eto</code> (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-filtre, -filter F</code>	filter: <code>B</code> , <code>V</code> , <code>R</code> (Johnson and Cousins), <code>Bc</code> , <code>Vc</code> , <code>Rc</code> , <code>Ha</code> , <code>OIII</code> , <code>g</code> , <code>r</code> , <code>i</code> , <code>z</code> , or the exact name (« Johnson R »); repeatable or comma-separated
<code>-nuit, -night YYYY-MM-DD</code>	night (evening date); repeatable
<code>-sans-dates-douteuses,</code>	leave out exposures with a shared DATE-OBS or dated in daytime
<code>-no-doubtful-dates</code>	
<code>-tout, -all</code>	the whole bank (≈ 78 GB)
<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
<code>-telechargements, -downloads N</code>	simultaneous downloads (automatic by default; at most 4)
<code>-debit, -rate MB/s</code>	maximum download rate (default 8 MB/s)
<code>-oui, -yes</code>	accept a selection over 5 GB without confirmation

coupole ohp anomalies (anomalies-report)

Duplicates (same file, daytime copy, identical pixels), shared or daytime dates, inconsistent field, classification to check, unknown instrument. Nothing is deleted.

<code>-dest FOLDER</code>	add the pixel duplicates found in this processing folder
<code>-csv FILE</code>	also write the list to this CSV file
<code>-nouveaux, -new</code>	only images new since the last refresh

coupole ohp classer (classify)

Without argument: list the names to check. With a database name and `-object`: record the correction (merge into an existing object if `-type` is omitted). `-online`: query JPL SBDB and CDS Sesame.

DB_NAME	target name as it appears in the database
<code>-objet, -object OBJECT</code>	canonical object to associate (existing one: merge)
<code>-type {ast,neocp,com,pla,tno,pn,neb,amas,gal,eto,autre}</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-oublier, -forget</code>	forget the correction of this name
<code>-en-ligne, -online</code>	resolve unknown names online (JPL SBDB, CDS Sesame), with cache

coupole ohp ranger (sort)

Useful after an interruption, or to merge several runs in the same folder.

<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
---------------------------	--

coupole ohp bilan (status)

progress of a processing folder

<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
<code>-json</code>	JSON output (for scripts)

coupole ohp fusionner (merge)

merge the state database on the share with the local working database (after a divergence)

<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
---------------------------	--

coupole ohp metadonnees (metadata)

For each XISF/FITS of the folder: FOCALLEN matched to the measured scale (old value in HISTORY) and the XISF properties Instrument:Sensor:XPixelSize, Instrument:Camera:XBinning, Instrument:Telescope:FocalLength... Pixels are not touched (read back and compared), atomic writing, log in `_traitement/metadonnees.csv`. Without `-rewrite`: dry run.

FOLDER	output folder (or one stack)
<code>-reecrire, -rewrite</code>	really write (otherwise: only say what would change); also records each file's location, relative to the output folder, in the state database (copy from an older processing run, another computer)
<code>-noms, -names {fr,en}</code>	language of folder names, object names and headers (default: interface language)

coupole archives liste (list)

list the known archives (stage, missions, status)

<code>-json</code>	JSON output (for scripts)
--------------------	---------------------------

coupole archives chercher (search)

Search MAST, ESO and IRSA (by default) for public final products around an object; `-archive` opus or pds with `-body` for planetary probes.

OBJECT	object (name or coordinates)
<code>-rayon, -radius ARCMIN</code>	search radius in arcminutes (default: 3)
<code>-archive A</code>	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
<code>-mission M</code>	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
<code>-instrument I</code>	instrument (part of the name, repeatable)
<code>-filtre, -filter F</code>	filter (exact name, repeatable)
<code>-debut, -from YYYY-MM-DD</code>	observations on or after this date
<code>-fin, -to YYYY-MM-DD</code>	observations up to this date
<code>-tous-niveaux, -all-levels</code>	also the calibrated single frames and the raw data
<code>-non-publiques, -proprietary</code>	also the still proprietary data (for information: not downloadable)
<code>-corps, -body BODY</code>	Solar System body (Voyager, Cassini, Juno)
<code>-limite, -limit N</code>	maximum number of rows per archive (default: 2000)
<code>-id ID</code>	keep only this identifier (repeatable)
<code>-json</code>	JSON output (for scripts)
<code>-csv FILE</code>	also write the results to this CSV file

coupole archives estimer (estimate)

Like « search », then add up the sizes (announced or measured) without downloading anything.

OBJECT	object (name or coordinates)
<code>-rayon, -radius ARCMIN</code>	search radius in arcminutes (default: 3)
<code>-archive A</code>	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
<code>-mission M</code>	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
<code>-instrument I</code>	instrument (part of the name, repeatable)
<code>-filtre, -filter F</code>	filter (exact name, repeatable)
<code>-debut, -from YYYY-MM-DD</code>	observations on or after this date
<code>-fin, -to YYYY-MM-DD</code>	observations up to this date
<code>-tous-niveaux, -all-levels</code>	also the calibrated single frames and the raw data
<code>-non-publiques, -proprietary</code>	also the still proprietary data (for information: not downloadable)
<code>-corps, -body BODY</code>	Solar System body (Voyager, Cassini, Juno)
<code>-limite, -limit N</code>	maximum number of rows per archive (default: 2000)
<code>-id ID</code>	keep only this identifier (repeatable)

coupole archives telecharger (download)

Like « search », then download (confirmation beyond the threshold: -yes), extract the science image and file it in <dest>/Archives.

OBJECT	object (name or coordinates)
-rayon, -radius ARCMIN	search radius in arcminutes (default: 3)
-archive A	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
-mission M	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
-instrument I	instrument (part of the name, repeatable)
-filtre, -filter F	filter (exact name, repeatable)
-debut, -from YYYY-MM-DD	observations on or after this date
-fin, -to YYYY-MM-DD	observations up to this date
-tous-niveaux, -all-levels	also the calibrated single frames and the raw data
-non-publiques, -proprietary	also the still proprietary data (for information: not downloadable)
-corps, -body BODY	Solar System body (Voyager, Cassini, Juno)
-limite, -limit N	maximum number of rows per archive (default: 2000)
-id ID	keep only this identifier (repeatable)
-dest FOLDER	output folder (default: the one of the Preferences)
-format {xisf,fits}	format of the prepared images (default: xisf)
-sans-original, -no-original	delete the original file once the image is prepared
-debit, -rate MB/S	rate cap in MB/s (default: setting, 8)
-paralleles, -parallel N	simultaneous downloads (default: 2, at most 6)
-max N	at most N files (the closest)
-oui, -yes	confirm a download beyond the threshold

coupole archives preparer (prepare)

Extract the science image of a multi-extension FITS, or convert a planetary product, to XISF or FITS Float32 next to the file.

FILE	files to prepare
-format {xisf,fits}	format of the prepared images (default: xisf)
-etiquette, -label FILE	detached PDS3 (.LBL) or PDS4 (.xml) label

coupole archives aligner (align)

Resample the images (WCS) onto the grid of one of them or an optimal grid; write the aligned files, the common mask, the colour preview and composition.txt.

FILE	prepared images (FITS or XISF with WCS)
-dest FOLDER	folder of the aligned files
-reference N	index of the reference image (default: 0, the first)
-optimale, -optimal	optimal grid covering every image
-methode, -method {bilineaire, adaptative, exacte, proche}	resampling: bilineaire (bilinear), adaptative (adaptive), exacte (exact, reproject), proche (nearest)
-echelle, -scale X	output pixel size factor (default: 1)
-sans-recadrage, -no-crop	do not crop to the common area
-format {xisf,fits}	format of the prepared images (default: xisf)

coupole archives bilan (status)

what is already downloaded and prepared

-dest FOLDER	output folder (default: the one of the Preferences)
-json	JSON output (for scripts)

coupole qualite

Optional: background, noise, FWHM and ellipticity (3×3 map), background gradient, saturation, trails, sampling

FOLDER	folder of stacks (or one image file)
-ecrire, -write	write QUALITE.csv and QUALITE.txt in each stack
-echantillon, -sample N	measure N images per stack (spread in time) instead of all
-tout, -all	measure everything even above 200 images (otherwise: sample of 5 per stack, announced)
-processus, -processes N	number of measuring processes (default: machine plan)
-avec-calibration,	also measure flats, darks, bias, masters and intermediate files (ex-
-with-calibration	cluded by default)

coupole donnees lire (read)

describe the contents of a file (FITS, CSV)

FILE	file to read
-json	JSON output (for scripts)

coupole donnees exporter (export)

export a spectrum or a series as CSV

FILE	file to read
-csv FILE	export a spectrum or a series as CSV
-hdu HDU	index of the dataset in the file (default: the first 1D one)
-vitesse, -velocity	add the radio velocity (frequency spectrum)
-f0 MHz	rest frequency in MHz (default: RESTFRQ, else H I)

coupole cosmo

From redshift to distances: comoving, luminosity, angular, lookback time, age, volume, distance modulus, scale (Planck 2018 and other parameter sets)

Z	redshift(s) to compute
-objet, -object NAME	object name: redshift requested from SIMBAD
-modele, -model {planck18,planck15, wmap9,simple,perso}	parameter set (default: planck18)
-h0 H0	H_0 in $\text{km s}^{-1} \text{Mpc}^{-1}$ (custom model)
-om OM	total Ω_m (custom model)
-ok OK	Ω_k , curvature (planck18 or perso)
-shoes	add the SH0ES comparison (planck18)
-sans-incertitudes,	without uncertainties (faster)
-no-uncertainties	
-csv FILE	write the results to this CSV file
-courbes, -curves ZMIN ZMAX N	write the curves (z from ZMIN to ZMAX, N points) to the CSV file given by -csv
-json	JSON output (for scripts)

coupole sites

Observing sites (OpenStreetMap map), time zones, UTC and local time

-ajouter, -add X	add a site: ID NAME LAT LON ALT TIMEZONE [MPC]
-supprimer, -delete ID	delete an added site
-heure, -time ISO_DATE	show a UTC date (ISO) in site time and evening date
-site ID	site identifier (default: ohp)

coupole machine

Hardware diagnosis, chosen parallelism, graphics card, ASTAP

-json	JSON output (for scripts)
-------	---------------------------

CHAPTER 14

Evolution

Sources. Addresses (TAP, collection, portal, download URL rewriting, resolvers, collection point, repository, map tiles) are in `coupole/donnees/sources.json`. Three layers: shipped values; file published in the repository, fetched at startup when newer and passing the shape check (known keys, types, domains from an allow-list; no cryptographic signature); values forced by the user, never overwritten. Preferences > Sources: edit, back to default, test. If the image server moves, `ohp.telechargement.de/vers` rewrites the URLs.

Classification. Shipped table `data/classement.json` (194 checked names); published aliases; user corrections (*Correct the classification*, `coupole ohp classify`); name-shape rules; optional online resolution (JPL SBDB, CDS Sesame), cached; grouping by position on a known fixed object; otherwise « unclassified ». Every automatic classification is marked « to check ».

Database. Extra ObsCore columns ignored, missing ones replaced by a default, unknown instruments and filters stored apart, non-image data never flagged « daytime »; tested on simulated inventories.

Security. No secret in the repository; remote sources file and collection point over HTTPS only, bounded size, domain allow-list; update archive downloaded from GitHub only and checked before extraction (absolute paths or « .. » refused, symbolic links refused, unpacked size and member count bounded), replacement with rollback on failure; XISF and FITS read with size bounds; folder names built from the database made safe (ASCII, forbidden characters, Windows reserved names, length); ASTAP launched without a shell; no `eval` nor `pickle` on external data.

CHAPTER 15

Incident reports and privacy

At first launch Coupole asks whether it may send anonymous reports: installation, sign of life (once a day), crash, native crash, frozen interface, written report. Content: version, system, processor, memory, graphics card, a random installation identifier, error trace with the home folder replaced by « ~ », the user name and the computer name masked and deep paths shortened; a report never exceeds 64 kB. Never a computer name, a user name or an image. Without consent nothing is sent: reports stay in `_rapports` in the settings folder. The choice can be changed (Help menu, `coupole reports -enable|-disable`).

CHAPTER 16

Writing a module

A module is a Python package with a `manifest.json` (identifier, version, French and English names, icon, texts, graphical and command-line entry points), discovered in `coupole/modules/`, through the `coupole.modules` entry point, or in the `modules/` folder of the settings. Texts through `tr()`, mandatory tooltips, long work off the GUI thread: see `CONTRIBUTING.md`.

CHAPTER 17

Troubleshooting

Symptom	What to do
Nothing starts (Windows) « Graphical interface unavailable »	Start menu > Coupole > Diagnostic: the error stays on screen. missing Qt libraries (Linux: <code>libxcb-cursor0</code> , <code>libegl1</code>); the command line still works.
Download fails	run again (resume); check the connection; Preferences > Sources > Test; server moved: URL rewriting.
« ASTAP not found »	ASTAP assistant; or <code>coupole astap -set PATH -catalog FOLDER</code> .
Incomplete catalogue	reinstall the catalogue (1,476 tiles expected).
Not enough space	choose another output folder; the space needed includes pending FITS.
Slow computer	economy mode, or fewer simultaneous conversions.
Unknown target name	Catalogue tab, « To check » box, <i>Correct the classification</i> .
Quality unavailable	<code>pip install sep</code> (included in the packages).
Settings back to defaults	the <code>reglages.json</code> file was unreadable: it is kept alongside with the suffix <code>.corrompu-...</code> .
Output folder not writable	clear message at start; choose another folder (Preferences or the Folder field).
File dialog without the NAS, or generic	Linux: install <code>xdg-desktop-portal</code> and the desktop backend (<code>-kde</code> , <code>-gnome</code> or <code>-gtk</code>), restart; KDE: <code>kio-fuse</code> ; otherwise type the path of the mounted share into the field (section 3.1).
System dialog does not open	0.1.10 and earlier, Spectra and series on KDE: update (nameless filter refused by the portal); otherwise Preferences > File dialogs: Qt; each opening is logged in <code>coupole.log</code> .
Owned images but nothing to open (copy from an older run)	update (0.1.11): locations are found through <code>journal.csv</code> (section 4.10).
Quality: empty columns on your own frames	update (0.1.11): tolerant XISF reader; the <i>status</i> column gives the reason.
<code>~/local/bin/coupole</code> not found	add <code>~/local/bin</code> to the PATH (line given by the installer).
<code>.part</code> or <code>.tmp</code> file in <code>_traitement</code>	left by a cut: resuming reuses it (<code>.part</code>) or replaces it; nothing to do.
WBPP does not find focal length and pixel	enter the values of the <i>For PixInsight / N.I.N.A.</i> box (Stacks tab) in the <i>Astrometric solution</i> panel; older files: <code>coupole ohp metadata FOLDER -rewrite</code> .
White image in N.I.N.A. or Siril	compatible XISF format (Processing tab), section 7.
« database is locked » towards a NAS (0.1.9 and earlier)	update: the local working database solves it (section 4.5).
« Two versions of the state database »	two computers (or the NAS) processed into the same folder: accept the merge, or <code>coupole ohp merge -dest FOLDER</code> ; both original databases are kept.
« Could not copy the state database to the share »	share unplugged or database open elsewhere: nothing is lost, retried on the next pass and at the next start.

Sources

« OHP student observations » bank, Observatoire de Paris / PADC (Paris Astronomical Data Centre), public TAP service, Etalab 2.0 licence; official guide of the database (PADC-BDD-DU-ECU). XISF 1.0 specification (Pleiades Astrophoto). ASTAP: <https://www.hnsky.org/astap.htm>. Observatory codes: Minor Planet Center. SEP: Barbary (2016), Bertin & Arnouts (1996). Adaptive moments: Bernstein & Jarvis (2002). Map background: © OpenStreetMap contributors. SIMBAD and Sesame: CDS, Strasbourg (Wenger et al. 2000); Aladin Lite (CDS); NED (NASA/IPAC); JPL Small-Body Database (NASA/JPL). Cosmology: Planck Collaboration (2020, A&A 641, A6); Hinshaw et al. (2013, WMAP 9); Riess et al. (2022, SH0ES); Hogg (1999, astro-ph/9905116); Komatsu et al. (2011) for the massive-neutrino density; astropy.cosmology; « cosmologie-redshift » calculator (same author). Archives: MAST (STScI), ESO Science Archive Facility, IRSA (NASA/IPAC), NOIRLab Astro Data Archive, Keck Observatory Archive (NExScI), SDSS, Planetary Data System (OPUS of the Ring-Moon Systems Node, Imaging Node); reproject (Astropy); Rector et al. (2007, AJ 133, 598) for chromatic ordering; amateur practices: see `docs/archives_methode.md`.